



НАЦИОНАЛЬНЫЙ ИССЛЕДОВАТЕЛЬСКИЙ
УНИВЕРСИТЕТ



ФАКУЛЬТЕТ ГЕОГРАФИИ
И ГЕОИНФОРМАЦИОННЫХ ТЕХНОЛОГИЙ

Курс «Моделировании климата городов» 2026, лекция №5

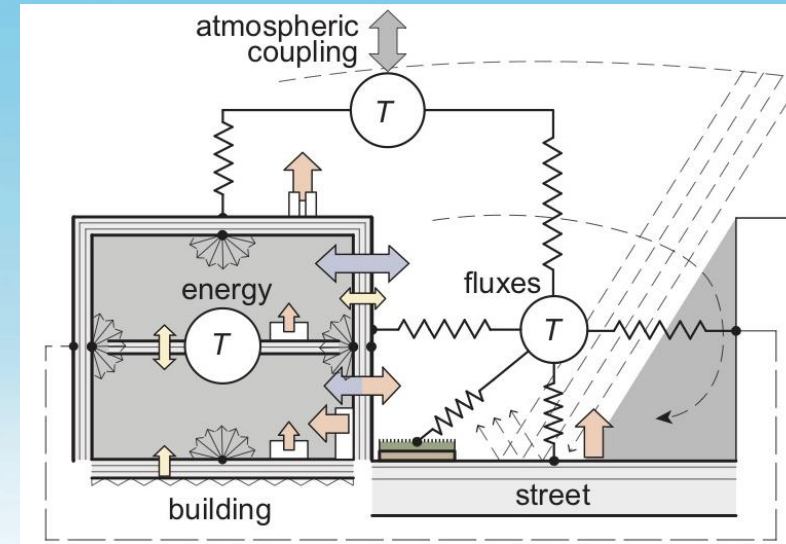
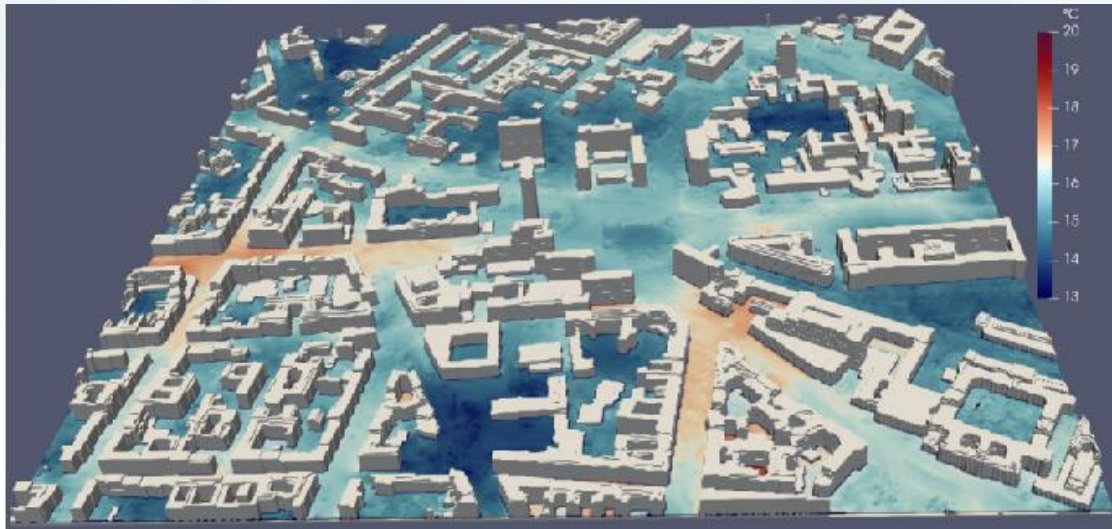
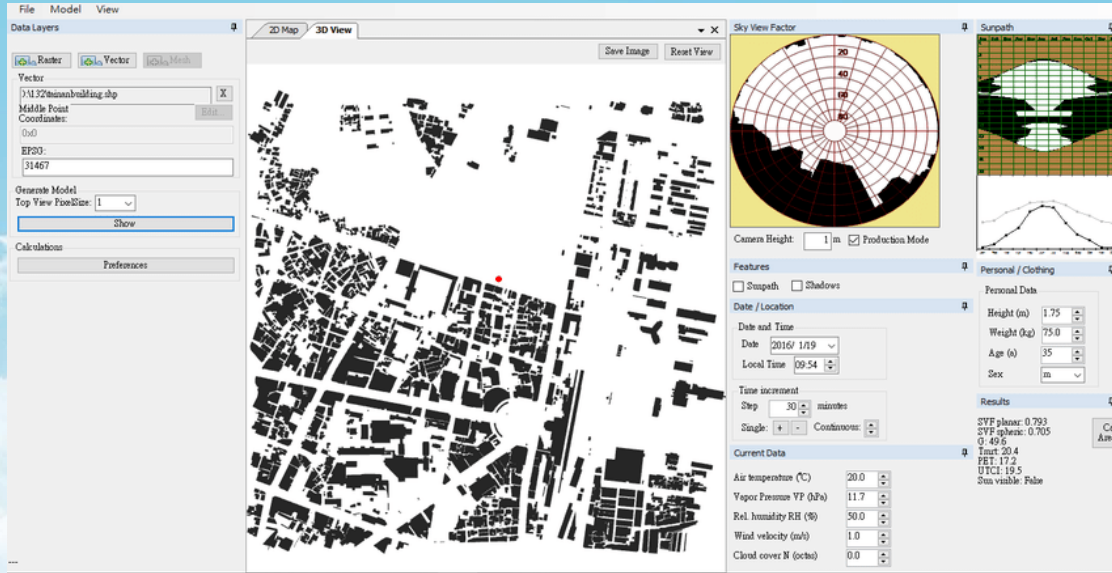
Пространственные данные для моделирования городского климата

Михаил Иванович Варенцов

mvarentsov@hse.ru

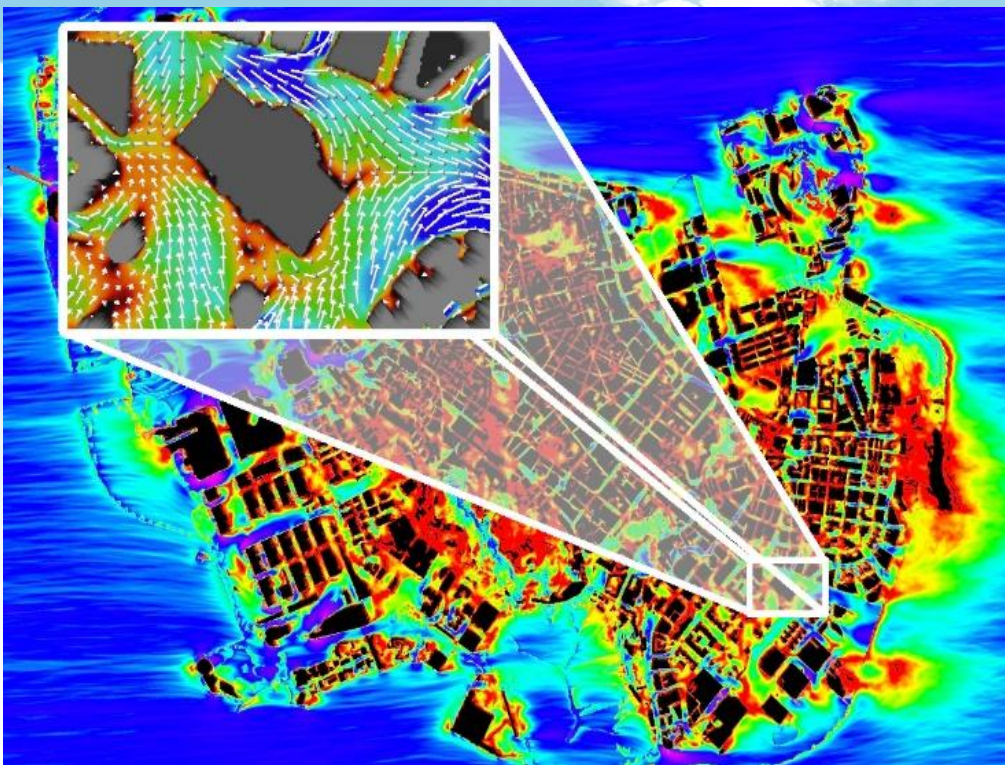


В предыдущих сериях...



В предыдущих сериях...

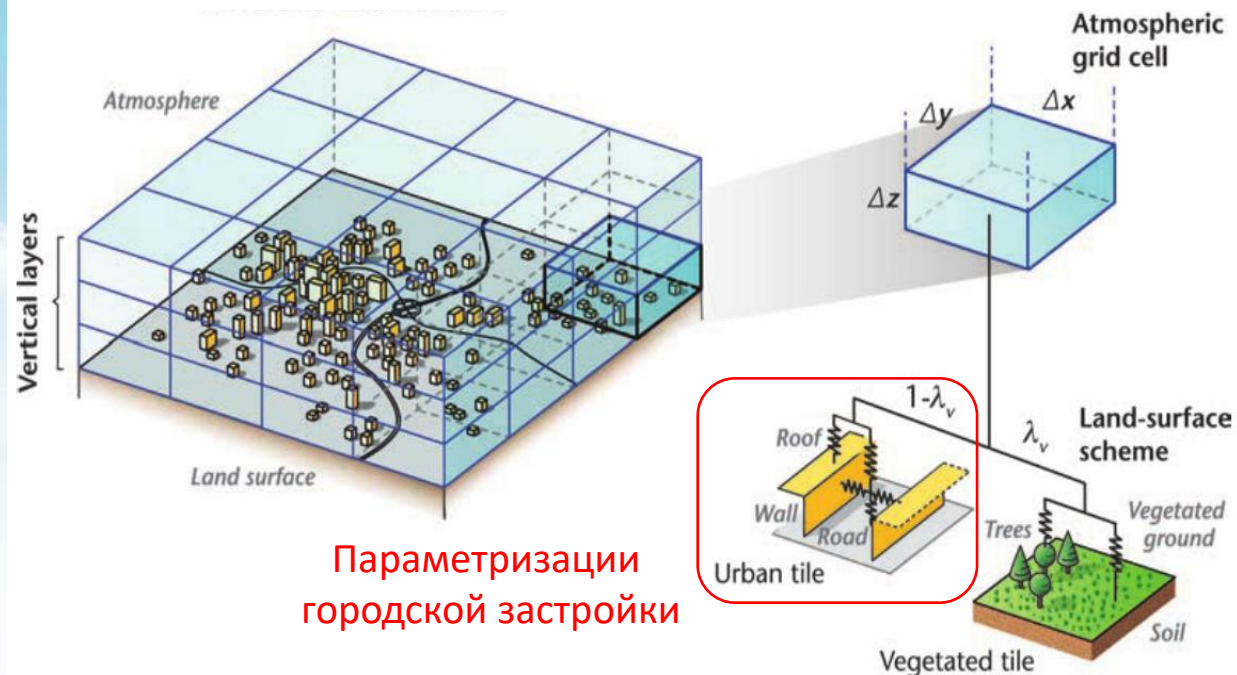
Микромасштабные модели (шаг сетки: первые метры)



ENVI _MET

Мезомасштабные модели

(шаг сетки: первые сотни метров – километры)



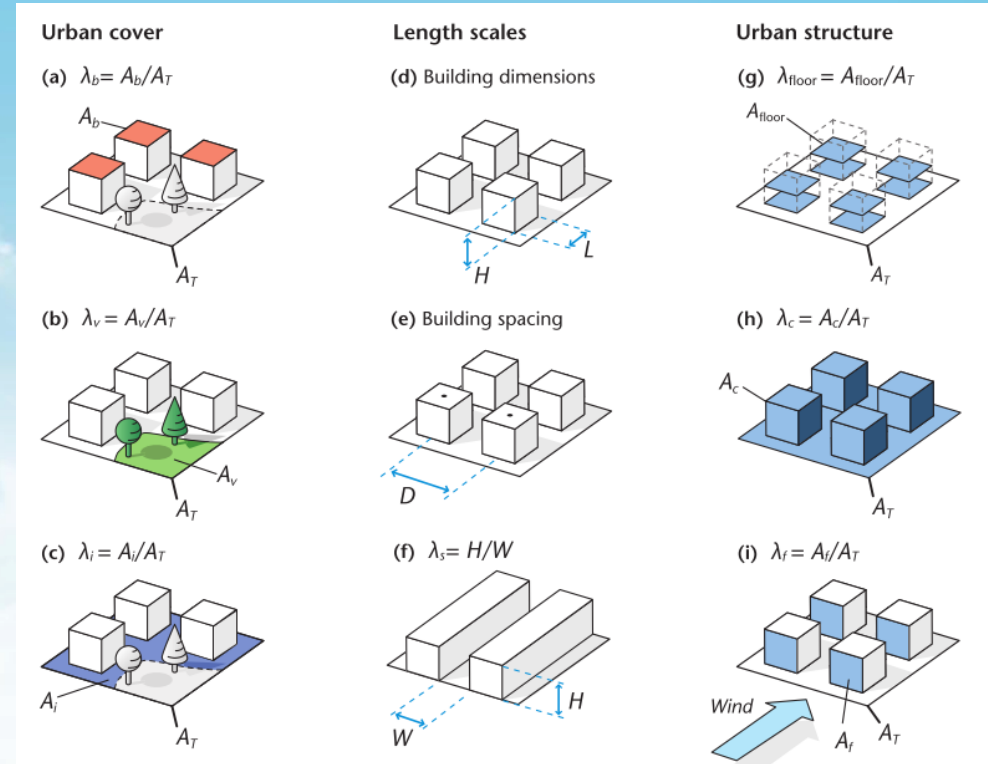
COSMO CONSORTIUM FOR SMALL SCALE MODELING



Внешние параметры городской среды

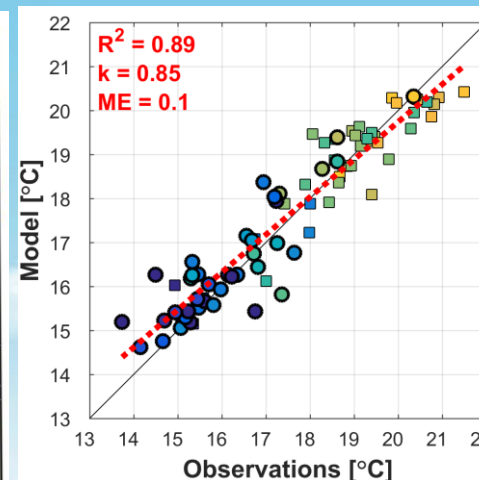
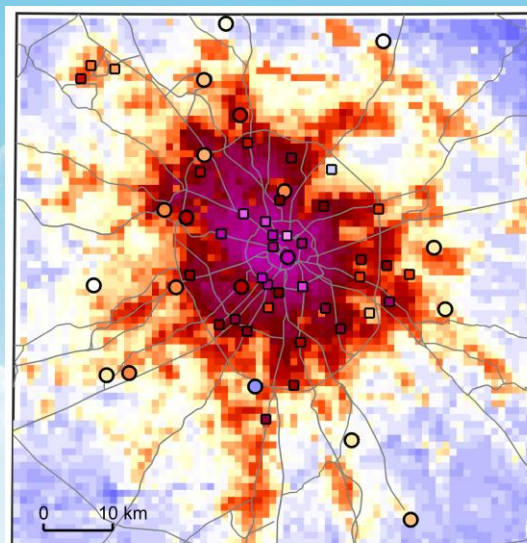
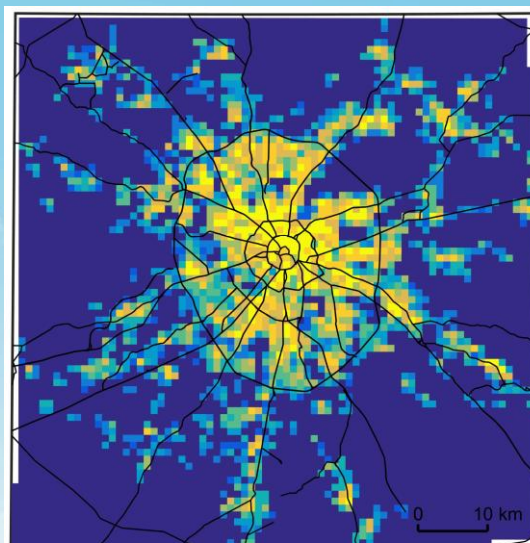
Какие параметры нужны?

- ❑ Данные о типах подстилающей поверхности:
где и расположены здания и другие искусственные объекты, водно-зеленая инфраструктура, какова их площадь?
- ❑ Геометрические параметры застройки:
какова форма и высота зданий?
- ❑ Теплофизические параметры городской среды (альbedo, теплоемкость, теплопроводность и пр.):
из каких материалов сделаны здания, дороги и пр.?
- ❑ Антропогенные эмиссии (тепла, влаги, загрязняющих веществ)

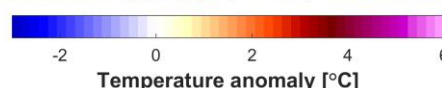
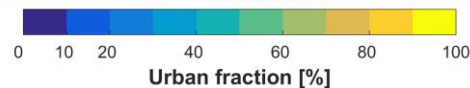
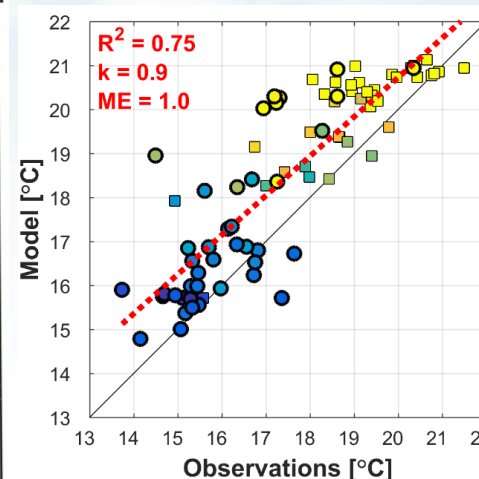
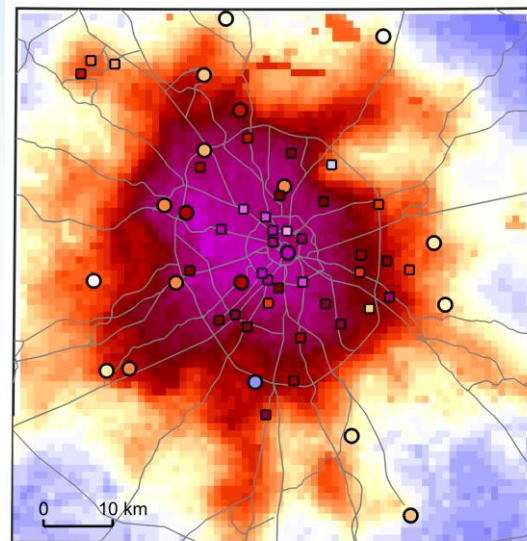
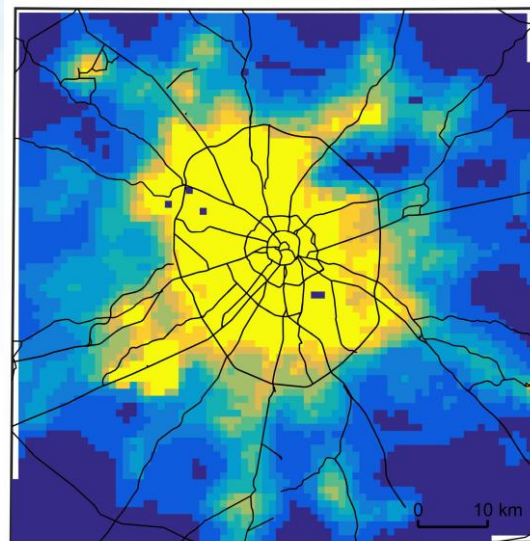


Плохие входные данные → плохие результаты

Реалистичная
конфигурация
застройки



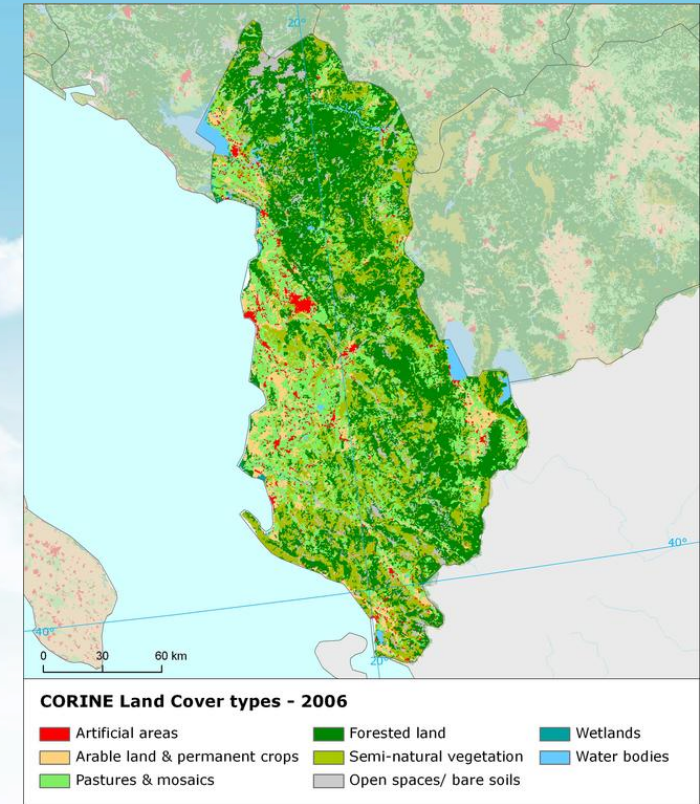
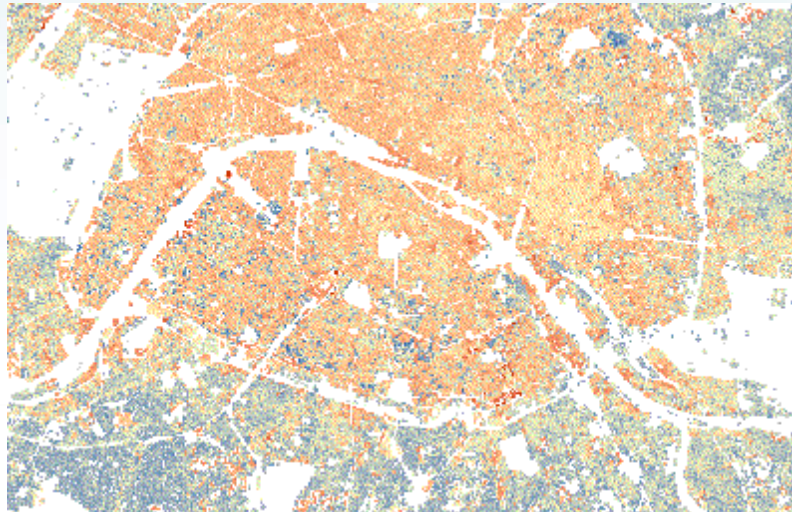
Данные по
умолчанию



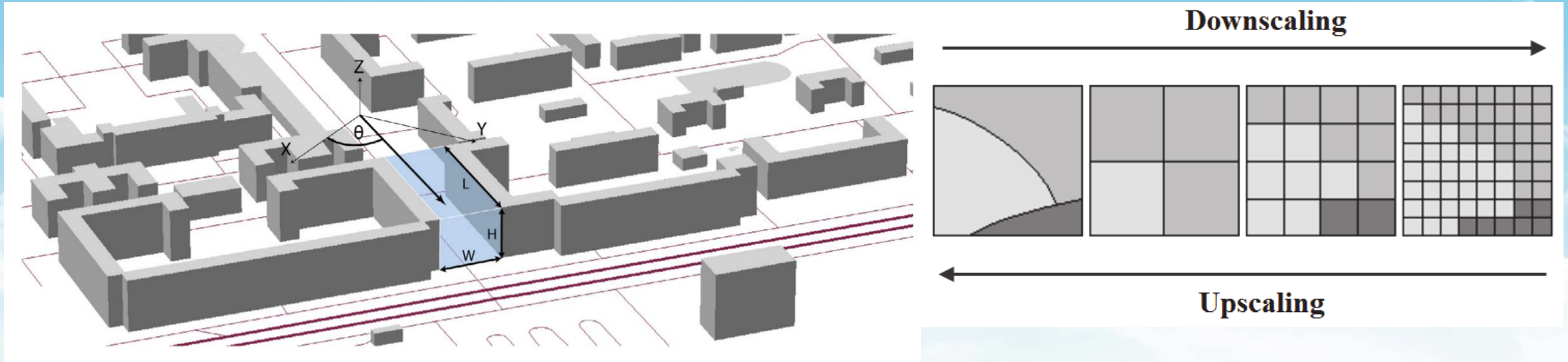
Ночная температура и ОТ за 5-25 августа
2017 (Varentsov et al., 2020)

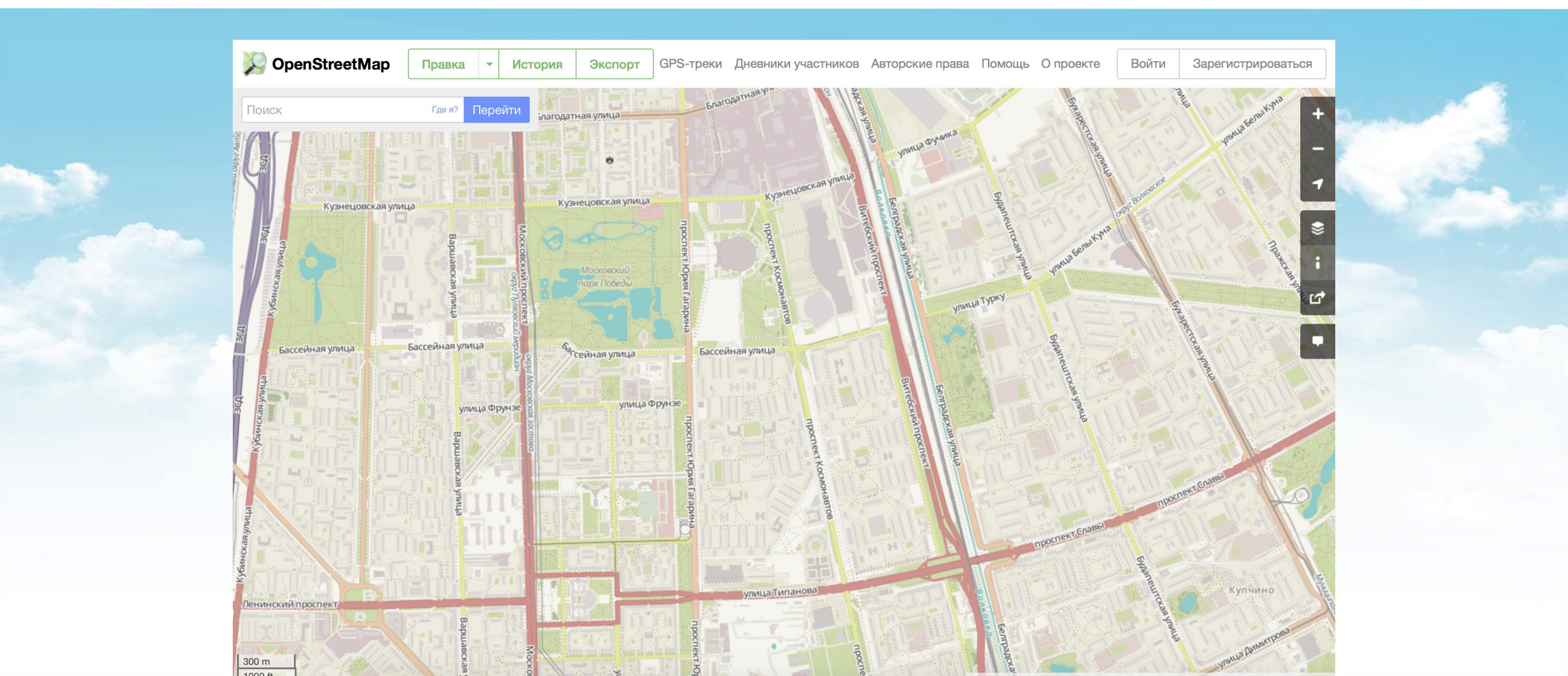
Источники пространственных данных

- ❑ Детализированные картографические данные в векторном формате (здания, дороги, зеленые насаждения и пр.)
 - Открытые свободно-распространяемые данные
 - Закрытые или частично открытые данные организаций и частных компаний (Геоцентр.Консалтинг, <https://digimap.ru/>; Яндекс.Карты, Google Maps, местные администрации и пр.)
- ❑ Глобальные/региональные базы данных о типах земельного покрова
- ❑ Глобальные/региональные специализированные базы данных о параметрах городской среды, необходимых для моделирования



Векторные картографические данные





<http://www.openstreetmap.org>

Данные OpenStreetMap

 **OpenStreetMap**

[Правка](#) [История](#) [Экспорт](#) [GPS-треки](#) [Дневники участников](#) [Авторские права](#) [Помощь](#) [О проекте](#)

 iamste



iamste

[Мои правки](#) 25 | [Мои заметки](#) | [Мои треки](#) 0 | [Мой дневник](#) 0 | [Мои комментарии](#) | [Мои настройки](#)

Зарегистрирован: 23 мая 2013 | Условия участия: Приняты больше 1 года назад

Аккаунт дает возможность вносить правки в карту и быть полноценным участником сообщества

Ваши друзья

Вы не добавили ещё ни одного друга.

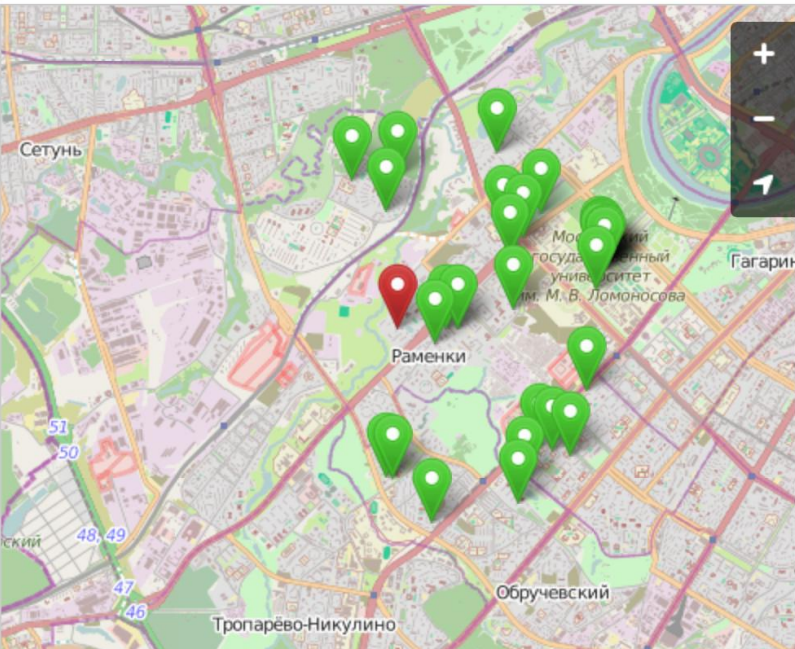
Другие ближайшие пользователи

[правки соседей](#) | [дневники соседей](#)

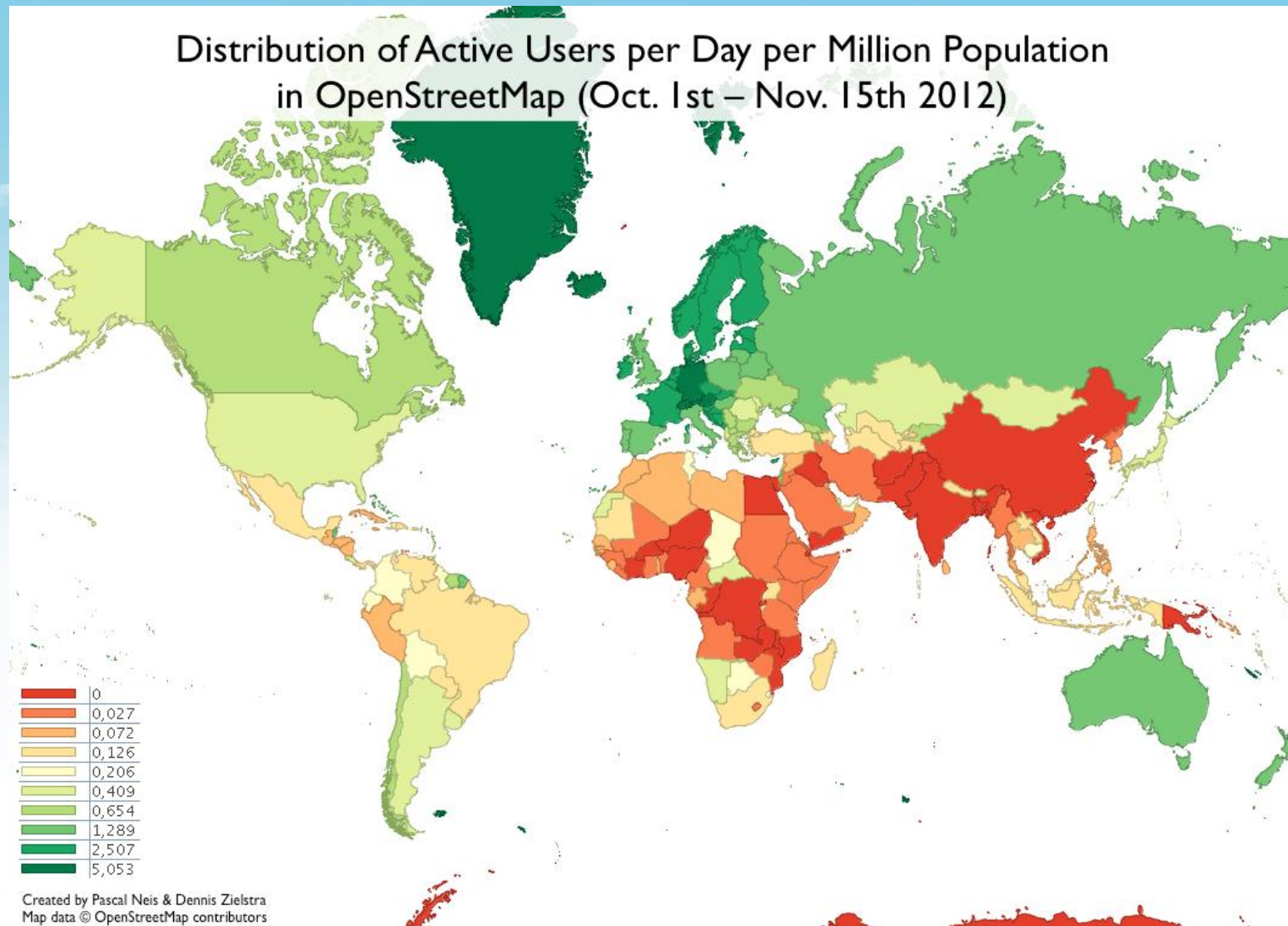
 **Renat A** (534 м от вас)
Последняя правка (больше 3 лет назад): "[Добавлена дорога к садоводству](#)"
[Отправить сообщение](#) | [Добавить в друзья](#)

 **maydond** (640 м от вас)
(нет правок)
[Отправить сообщение](#) | [Добавить в друзья](#)

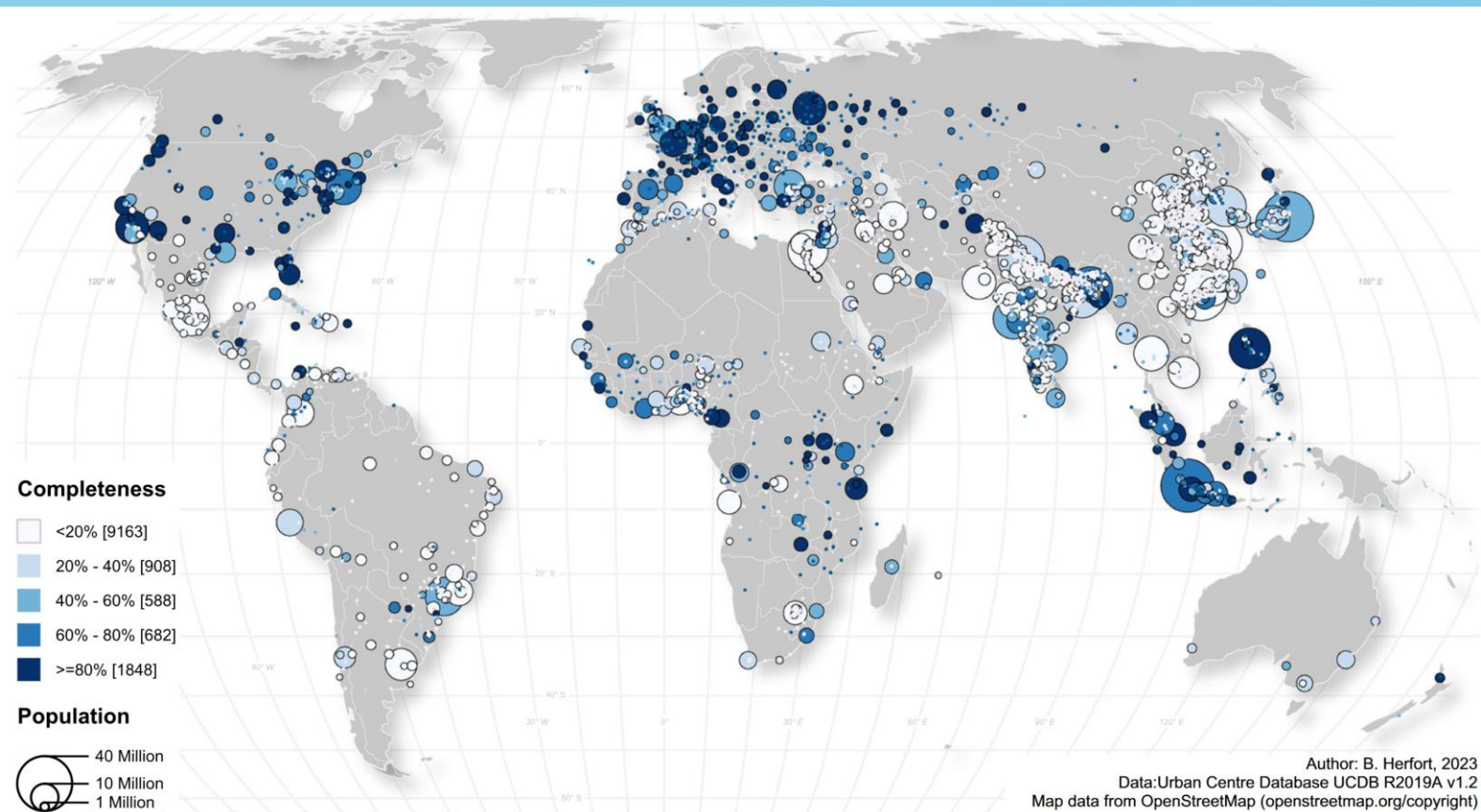
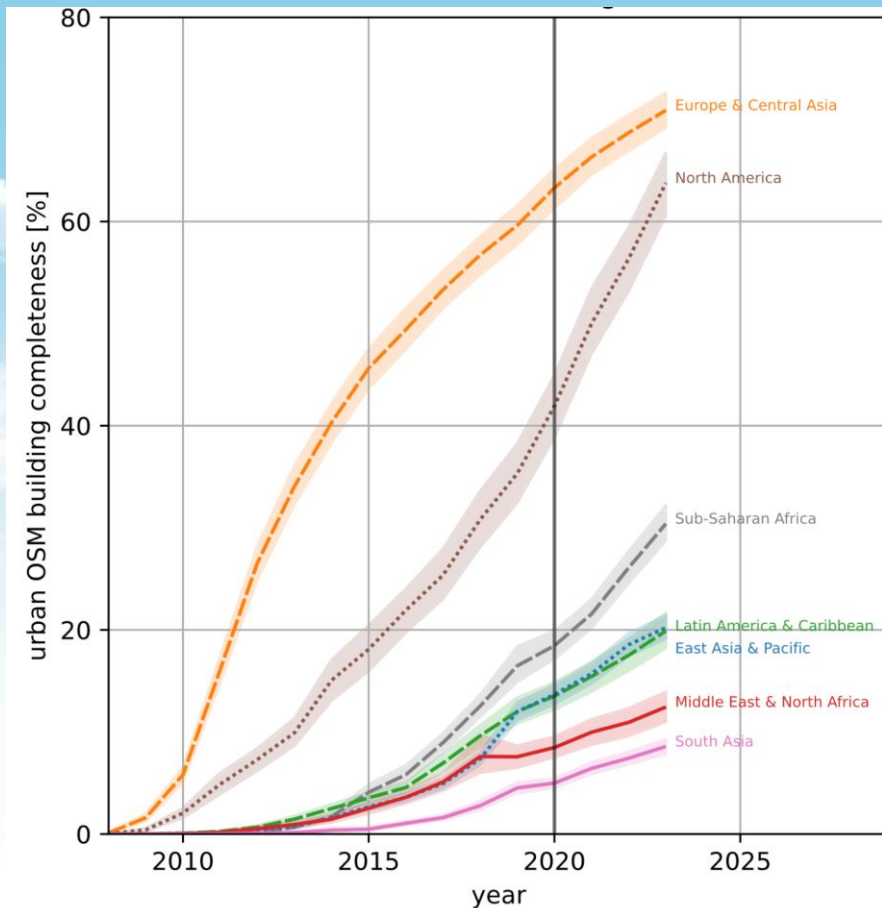
 **parshin** (790 м от вас)
Последняя правка (11 месяцев назад): "[По результатам ПВД](#)"
[Отправить сообщение](#) | [Добавить в друзья](#)



Данные OpenStreetMap



Данные OpenStreetMap



Herfort et al. (2023). A spatio-temporal analysis investigating completeness and inequalities of global urban building data in OpenStreetMap. *Nat Commun* 14, 3985. doi: [10.1038/s41467-023-39698-6](https://doi.org/10.1038/s41467-023-39698-6)

Данные OpenStreetMap

Здания

Key	Value
building	yes



- Тег **building = yes** означает, что объект является зданием и его следует интерпретировать как полигональный объект
- Это минимальная форма, которая используется, когда точно не известен тип здания, или этот тип отмечен отдельным тегом **building:type=***
- Примеры:
 - **building = apartments** многоквартирный жилой дом,
 - **building = hospital** больница,
 - **building = industrial** промышленное строение,

В OSM здания - это, в том числе, *подземные сооружения* с помещениями.

К зданиям относятся, в том числе (но не исключительно): жилые здания, промышленные здания (производственные цеха, складские здания), сельскохозяйственные постройки для хранения продукции, материалов и содержания животных

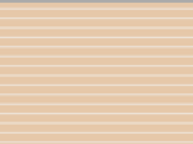
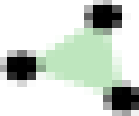
Атрибуты зданий

entrance	yes / main / exit / service / emergency	Вход в здание.
height	number	Высота в метрах
building:levels	number	Этажность
building:fireproof	yes/no	Информация об огнестойкости при пожаре.
building:levels	number	Этажность
min_level	number	Минимальный этаж
max_level	number	Максимальный этаж (обычно не используется)

Данные OpenStreetMap

Типы землепользования

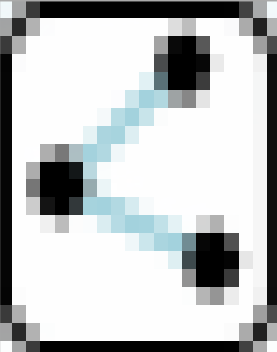
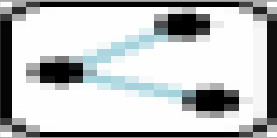
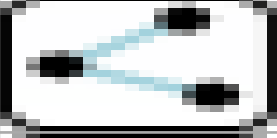
Землепользование кодируется с помощью тега **landuse=***

Ключ	Значение	Элементы	Описание	Обозначение	Фото
landuse	allotments		Земельные участки, предоставленные местным жителям для выращивания овощей и цветов. Можно использовать для обозначения садоводческих товариществ , приусадебных участков, огородов . Allotment (gardening)		
landuse	basin		Бассейны стока		
landuse	brownfield		Расчищенные от строений районы под новую застройку		
landuse	cemetery		Кладбища. <code>religion="religion"</code> религия захоронения (список здесь) Малые захоронения можно указывать как <code>amenity="grave_yard"</code>		

Данные OpenStreetMap

surface=*

Тег surface используется для более подробного описания покрытия. Этот тег может использоваться с теми объектами, где поверхность покрыта каким-либо материалом.

Ключ	Значение	Элементы	Описание	Фото
surface	paved		Этот тег ставится на линию, вместе тегом (highway), если дорога, которую он описывает, в основном имеет твёрдое (не сыпучее) покрытие по всей длине, например покрыта битумом, асфальтом или вымощена камнем (<i>Это общее, грубое описание поверхности</i>).	
surface	asphalt		Асфальтобетон , обычно называемый асфальтом .	
surface	cobblestone		Булыжник. Специально уложенные относительно ровными поверхностями вверх камни. К булыжникам не применяют дополнительную обработку.	
surface	cobblestone:flattened		То же, что и брусчатка surface=sett	

Данные OpenStreetMap

Как и где скачать?

The screenshot displays the OpenStreetMap Export Tool interface. The left sidebar shows 'File Formats' with options like Shapefile (.shp), GeoPackage (.gpkg), Garmin (.img), Google Earth (.kml), OSM (.pbf), MAPS.ME (.mwm), OsmAnd (.obf), and MBTiles (.mbtiles). The main area is divided into four tabs: '1 Describe', '2 Formats', '3 Data', and '4 Summary'. The '3 Data' tab is active, showing a 'Tag Tree' configuration. A red box highlights the 'Tag Tree' tab, and another red box highlights the 'Communication' category, which is expanded to show 'Geometry types: point, polygon' and 'Keys: name, office, man_made, tower, operator, communication:mobile, communication:radio'. A red text overlay 'Выбор по дереву тегов' (Selection by tag tree) points to the 'Tag Tree' tab. Another red text overlay 'Стандартная конфигурация' (Standard configuration) points to the 'Communication' category. The right side of the interface shows a map of Moscow with a blue dashed box indicating the 'Area Of Interest (AOI)' and a search bar. The bottom of the interface includes a 'Contact Us' link, a 'Made with ❤️ by HOT and friends' note, and a 'Fork the Code' link.

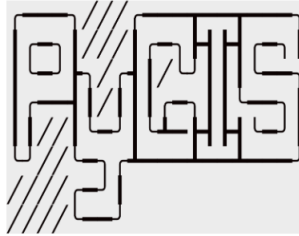
Выбор по дереву тегов

Стандартная конфигурация

<https://export.hotosm.org/>

Данные OpenStreetMap

Как и где скачать?



Python Open Source
Spatial Programming &
Remote Sensing

🔍 Search this book...

PyGIS - Open Source Spatial
Programming & Remote Sensing

0 - GET STARTED IN SPATIAL PYTHON

Welcome - Let's get started

Getting Started in Python

Setting up a Normal Python
Environment

Geospatial Environment Installation
Guide

An Introductory Example

Learn More



Download and visualize OpenStreetMap data with OSMnx

One the most useful features that OSMnx provides is an easy-to-use way of retrieving [OpenStreetMap](#) data (using [OverPass API](#)).

In this tutorial, we will learn how to download and visualize OSM data covering a specified area of interest: the neighborhood of Edgewood in Washington DC USA.

```
# Specify the name that is used to search for the data
place_name = "Edgewood Washington, DC, USA"
```

OSM Location Boundary

Let's also plot the Polygon that represents the boundary of our area of interest (Washington DC). We can retrieve the Polygon geometry using the `ox.geocode_to_gdf` [docs]

([#https://osmnx.readthedocs.io/en/stable/osmnx.html?](https://osmnx.readthedocs.io/en/stable/osmnx.html?highlight=geocode_to_gdf)

[#https://osmnx.readthedocs.io/en/stable/osmnx.html?highlight=geocode_to_gdf](https://osmnx.readthedocs.io/en/stable/osmnx.html?highlight=geocode_to_gdf) function.

```
# import osmnx
import osmnx as ox
import geopandas as gpd

# Get place boundary related to the place name as a geodataframe
area = ox.geocode_to_gdf(place_name)
```

As the name of the function already tells us, `gdf_from_place()` returns a GeoDataFrame based on the specified place name query.

https://pygis.io/docs/d_access_osm.html

OSMnx



OpenStreetMap

Implementation in Python

Data Science
PAPER

Данные Overture Maps



Collaborative Map Building



Global Entity Reference System



Structured Data Schema



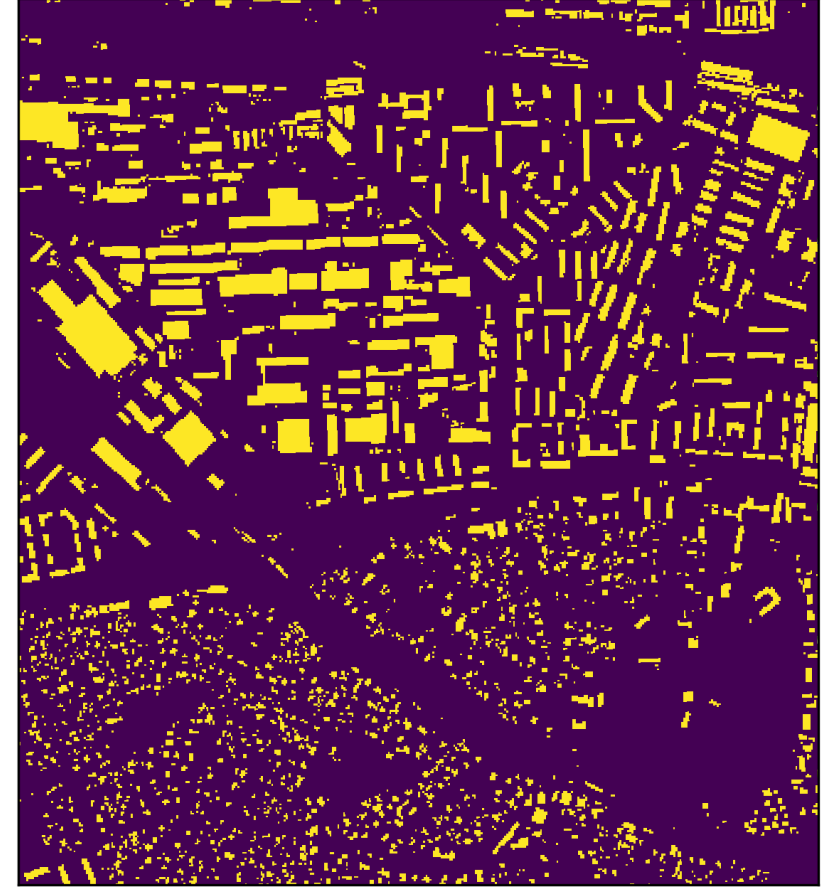
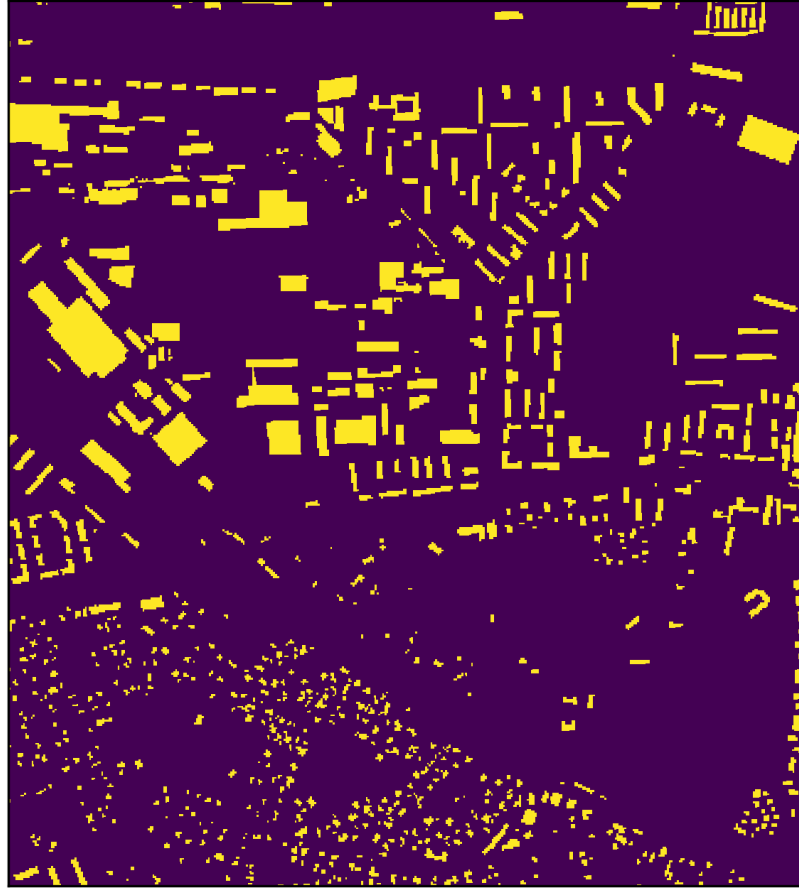
Quality Assurance Processes

Автор слайда: Окунева В.В., кафедра картографии и геоинформатики географического факультета МГУ



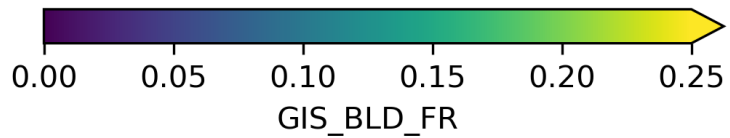
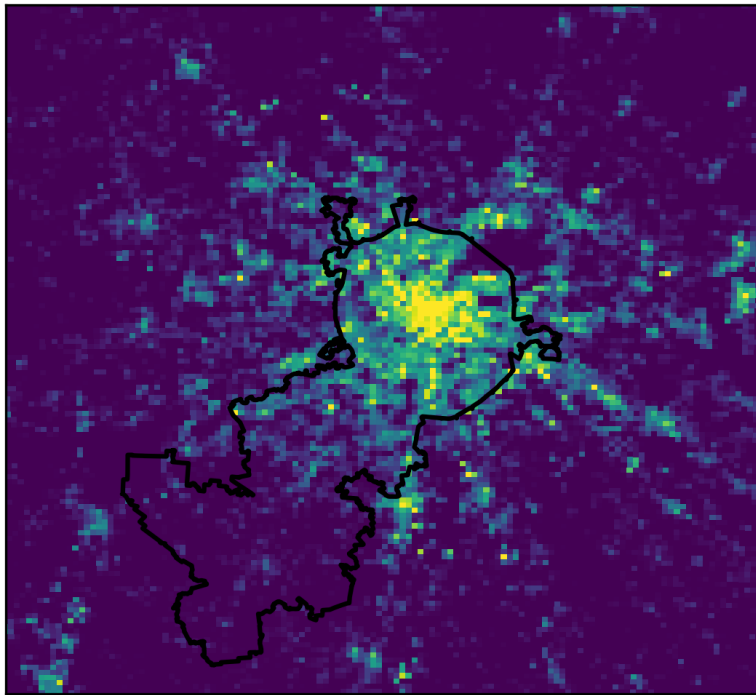
- адреса "*addresses*"
- основа "*base*"
- здания "*buildings*"
- границы "*divisions*"
- места "*places*"
- транспорт "*transportation*"

Overture Maps vs OpenStreetMap

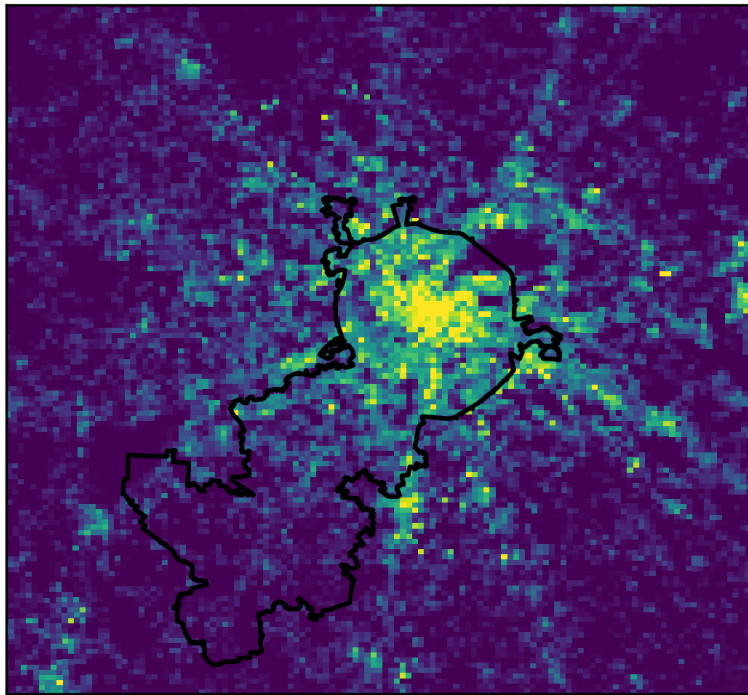


Данные Overture Maps

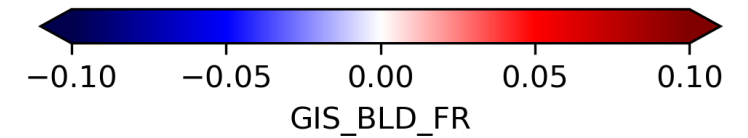
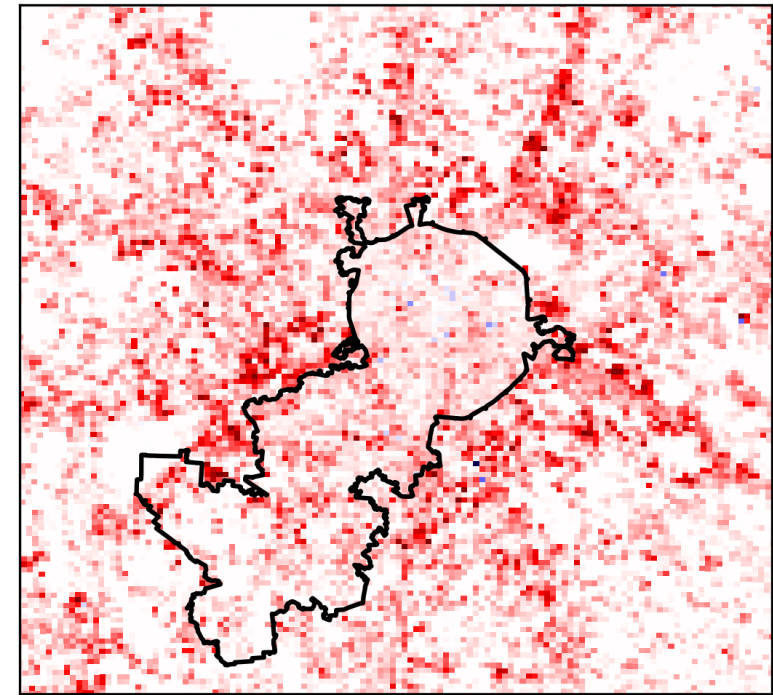
OSM



OVM



OVM - OSM



Данные Overture Maps

column_name	column_type	description
id	string	an Overture feature's unique id, part of the Global Entity Reference System (GERS)
geometry	binary	well-known binary (WKB) representation of the feature geometry
bbox	struct<xmin: float, xmax: float, ymin: float, ymax: float>	area defined by two longitudes and two latitudes: latitude is a decimal number between -90.0 and 90.0; longitude is a decimal number between -180.0 and 180.0.
theme	string	one of six Overture data themes
type	string	one of 14 Overture feature types
version	int32	version number of the feature, incremented in each Overture release where the geometry or attributes of this feature changed
sources	list<element: struct<property: string, dataset: string, record_id: string, update_time: string, confidence: double>>	array of source information for the properties of a given feature

Пример хранения атрибутивной информации

Overture Maps имеет свою выработанную стандартизированную систему хранения различных атрибутов.

Так, все данные представлены в формате *GeoParquet*, а структура таблиц атрибутов соответствует *JSON* — схеме.

Колоночно-ориентированный формат *GeoParquet*, позволяет обрабатывать большие объемы данных по отдельным необходимым столбцам независимо друг от друга.

Регулярный выпуск обновлений позволяет опираться на *Overture Maps* в долгосрочной перспективе.

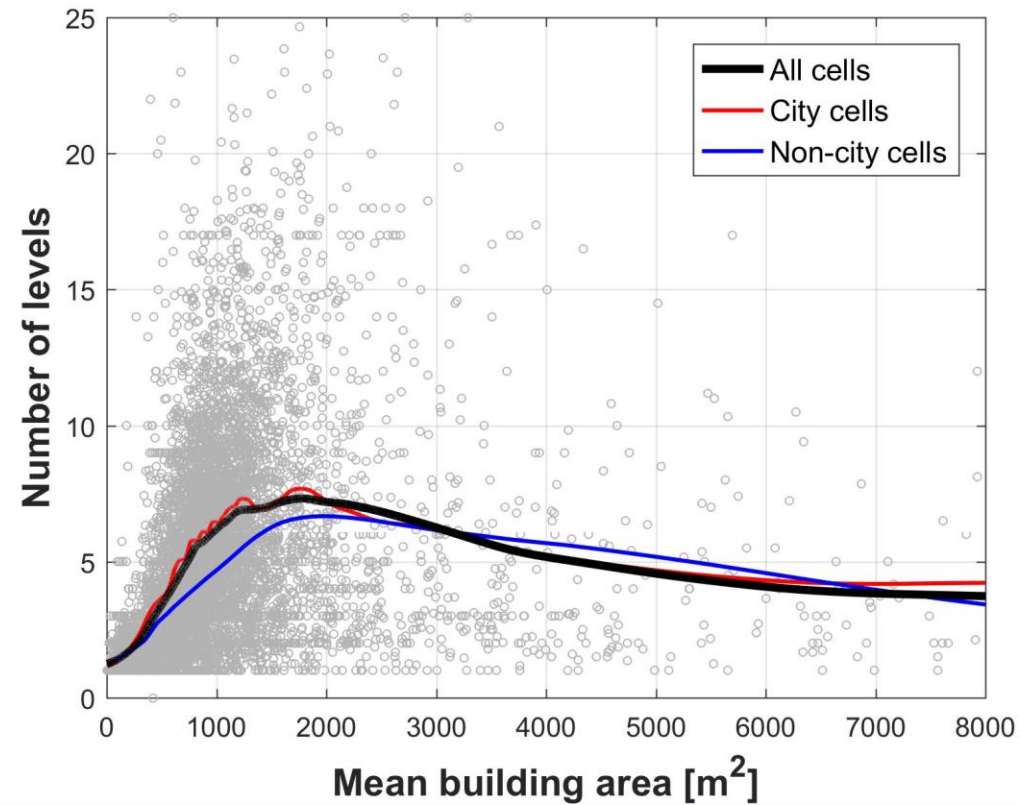
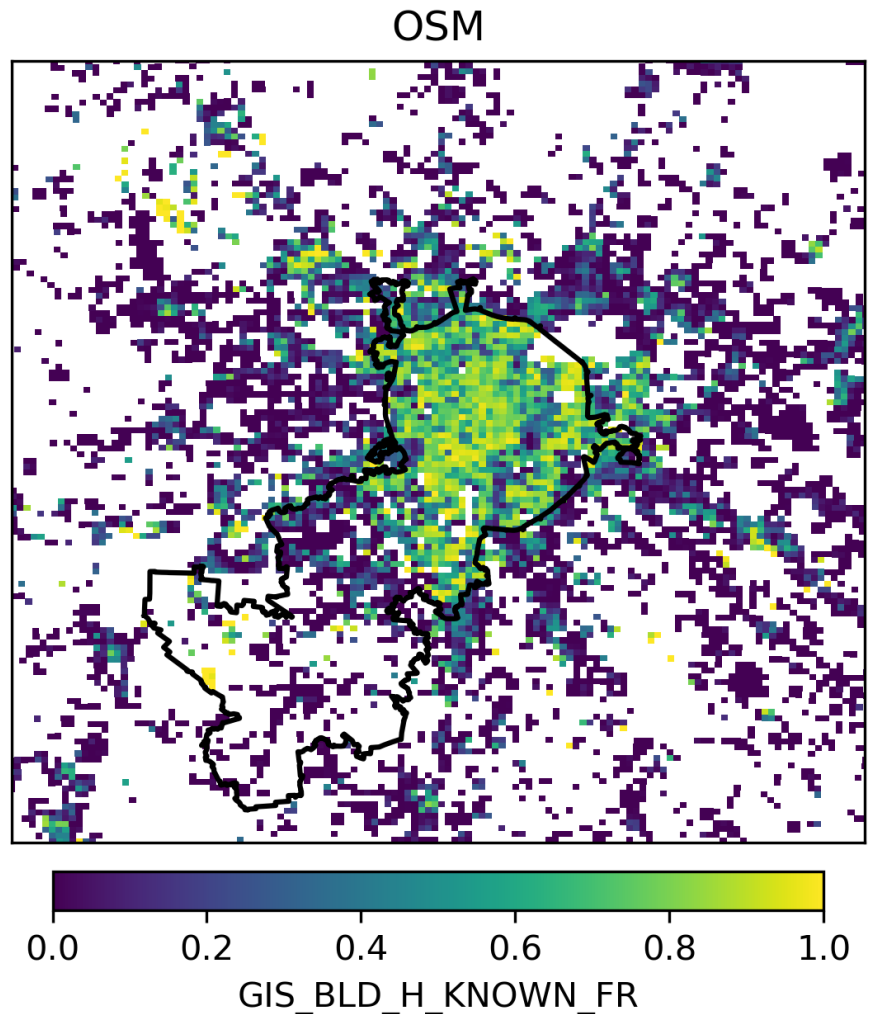
Работа с Overture Maps

- Полный объем данных в формате *GeoParquet* через ресурсы *Amazon S3* и *Microsoft Azure Blob Storage* (недоступен на территории России)
- *DuckDB*
- Через командную строку *Python*
(<https://github.com/OvertureMaps/overturemaps-py>)
- Данные небольшого охвата инструментом [Overture's Explore tool](#)

```
$ pip install overturemaps
```

```
$ overturemaps download --bbox=-71.068,42.353,-71.058,42.363 \  
-f geoparquet --type=building --output=boston.geoparquet
```


Проблема неполноты данных о высоте зданий



Самсонов, Варенцов, 2020 (Метеорология и гидрология)

Решение в подходе машинного обучения

Table 1. Predictors used in ML model

Type of predictor	Specific predictor	Name
Building's functions	Type of the building (can include several classes)	subtype
	Class of the building (a narrower concept than type)	class
Morphological properties of building footprint	Footprint area	area
	Aspect ratio (height/width) of the minimum bounding rectangle	obox_hw
	Compactness, proximity of the building's shape to a circle (ratio of the building area to the area of a circle having the same perimeter)	isquoton
	Compactness, proximity of the building's shape to a rectangle (ratio of the area of the minimum bounding rectangle to the area of the building)	obox_ratio
Properties of building's surroundings	Average width of the urban canyons surrounding the building	width_med
	The prevailing LCZ type in the 100-m radius	lcz_majori
The building height extracted from ArcticDEM	Alternative estimate of building height	min_dem

Okuneva et al., 2025, submitted

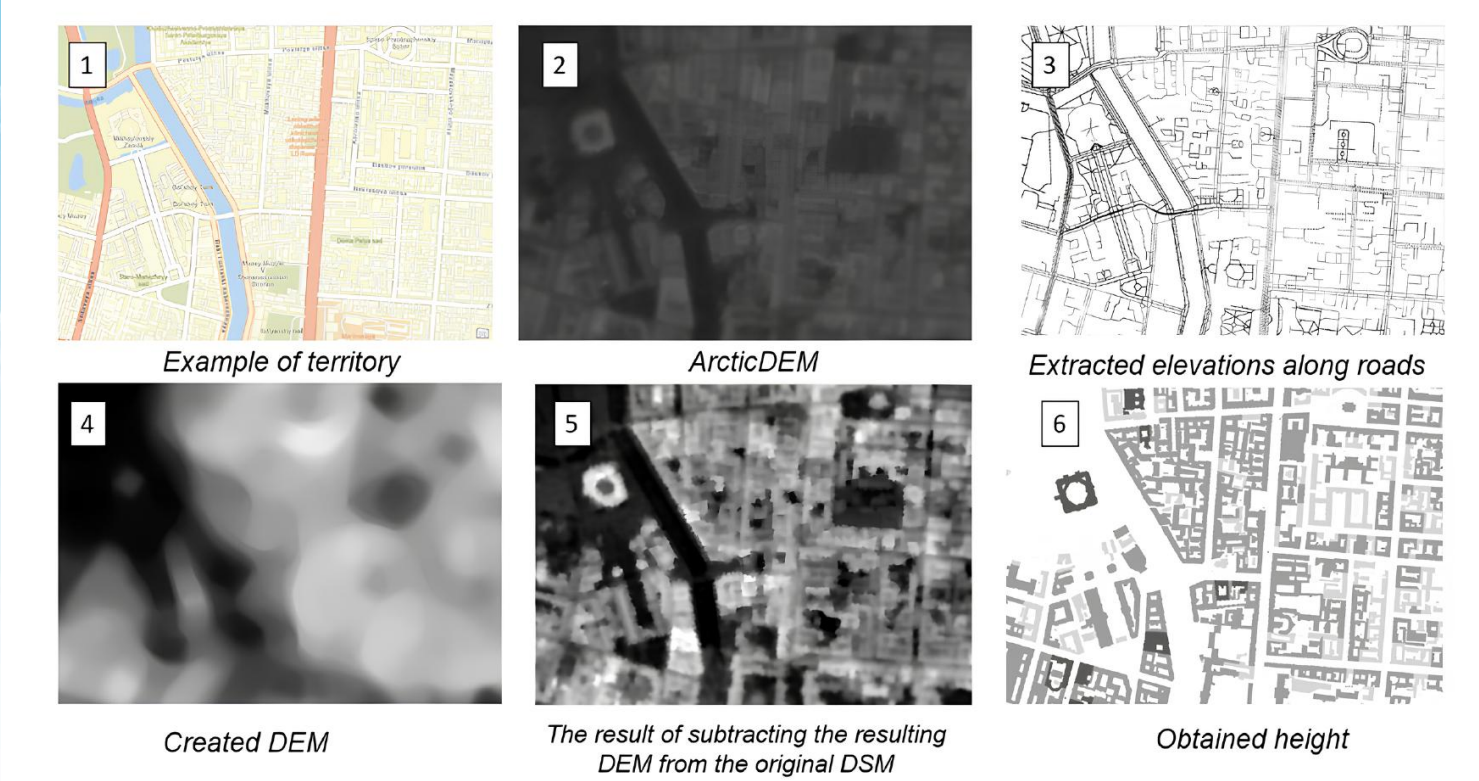


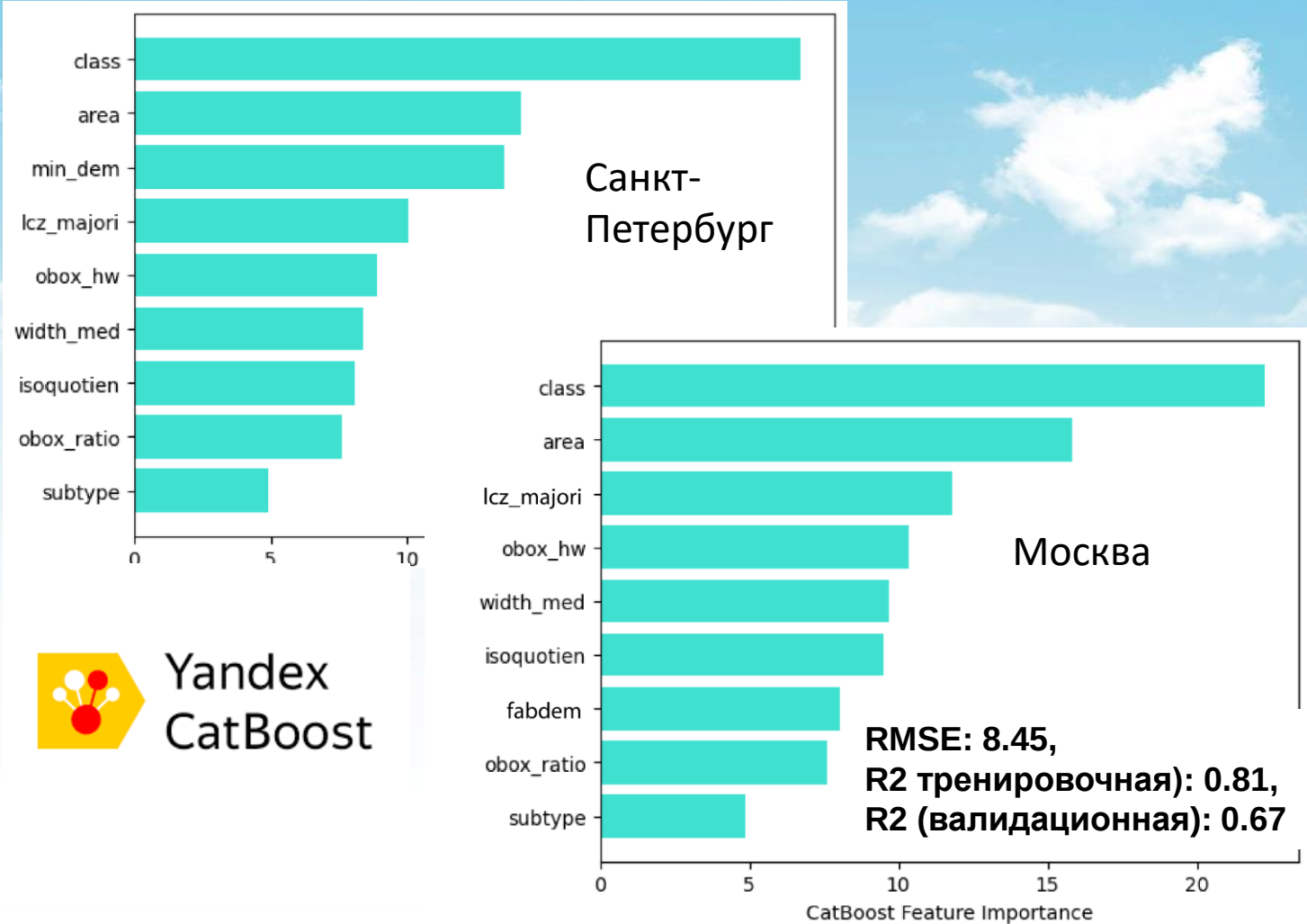
Figure 1. Steps of the extraction building height from ArcticDEM

Решение в подходе машинного обучения

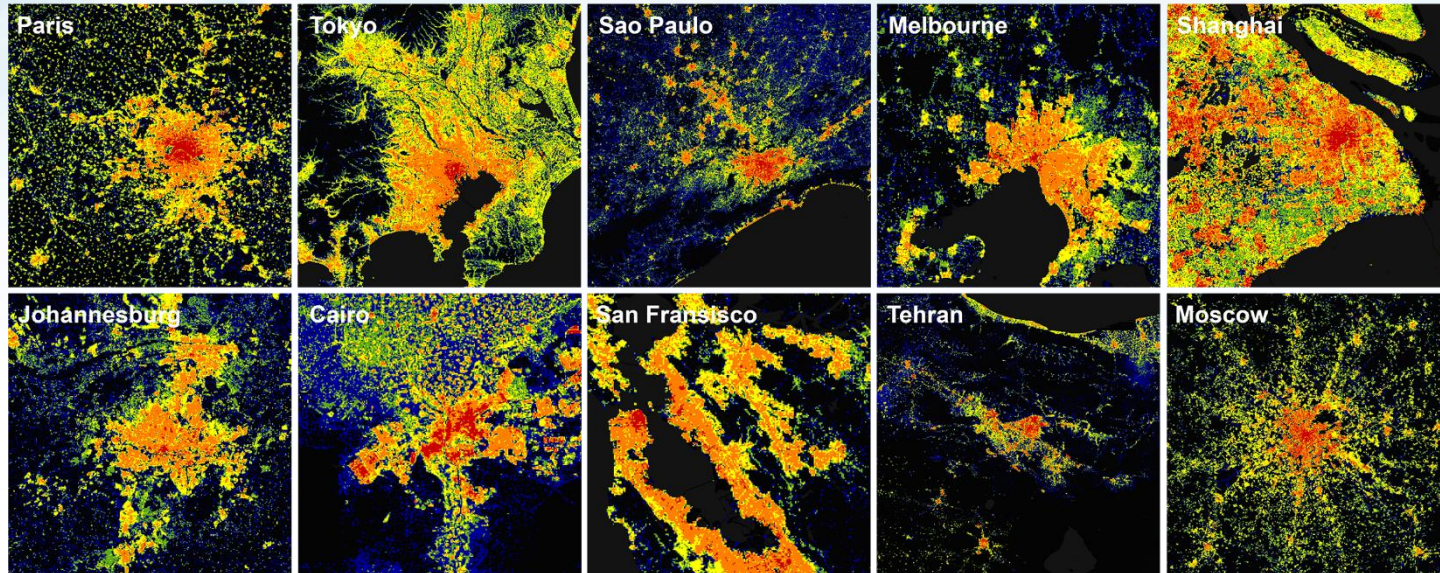
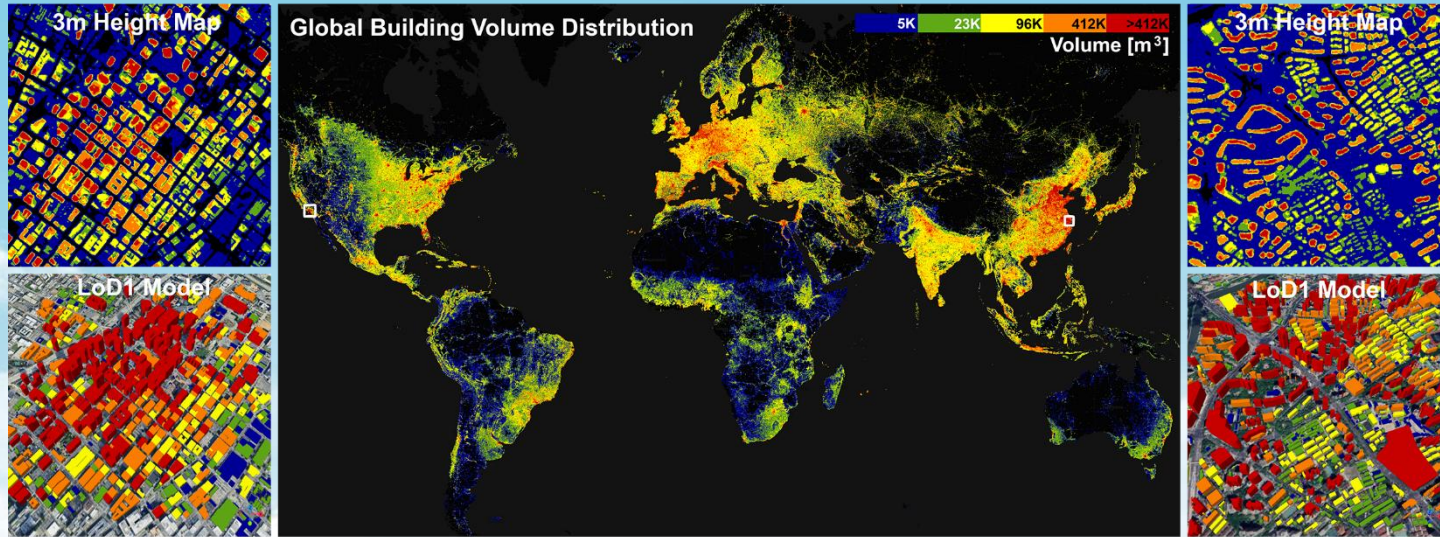
Table 1. Predictors used in ML model

Type of predictor	Specific predictor	Name
Building's functions	Type of the building (can include several classes)	subtype
	Class of the building (a narrower concept than type)	class
Morphological properties of building footprint	Footprint area	area
	Aspect ratio (height/width) of the minimum bounding rectangle	obox_hw
	Compactness, proximity of the building's shape to a circle (ratio of the building area to the area of a circle having the same perimeter)	isoquotien
	Compactness, proximity of the building's shape to a rectangle (ratio of the area of the minimum bounding rectangle to the area of the building)	obox_ratio
Properties of building's surroundings	Average width of the urban canyons surrounding the building	width_med
	The prevailing LCZ type in the 100-m radius	lcz_majori
The building height extracted from ArcticDEM	Alternative estimate of building height	min_dem

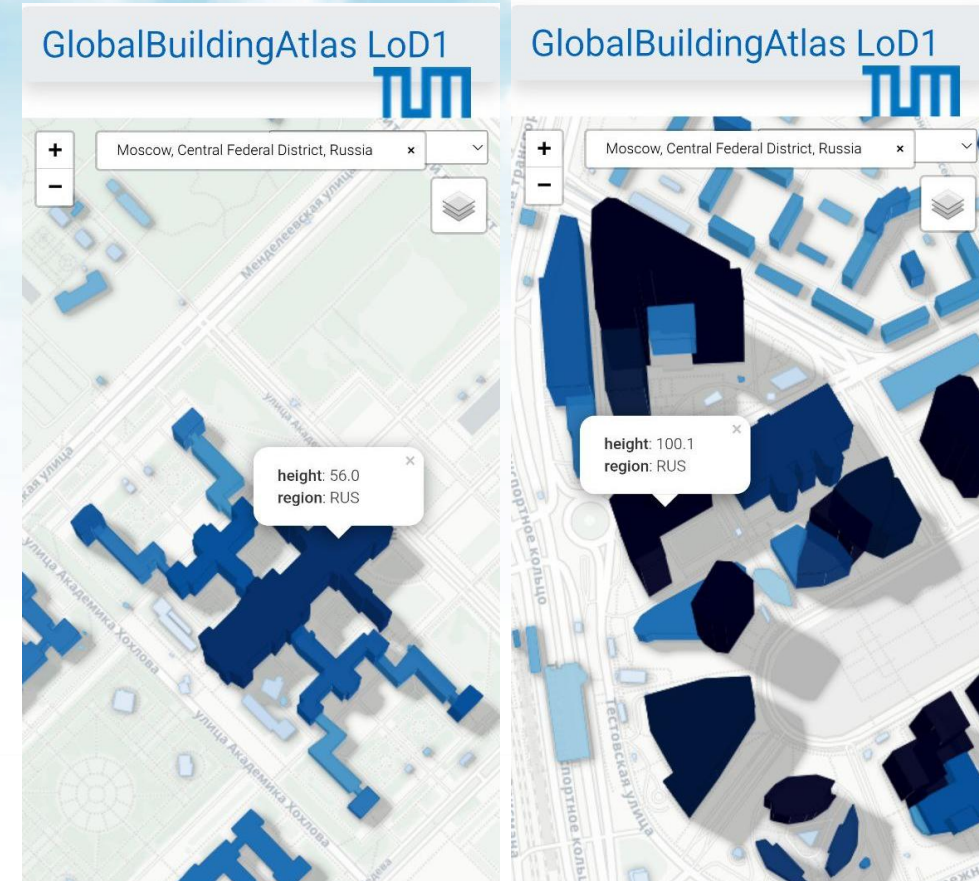
Okuneva et al., 2025, submitted



Global Building Atlas



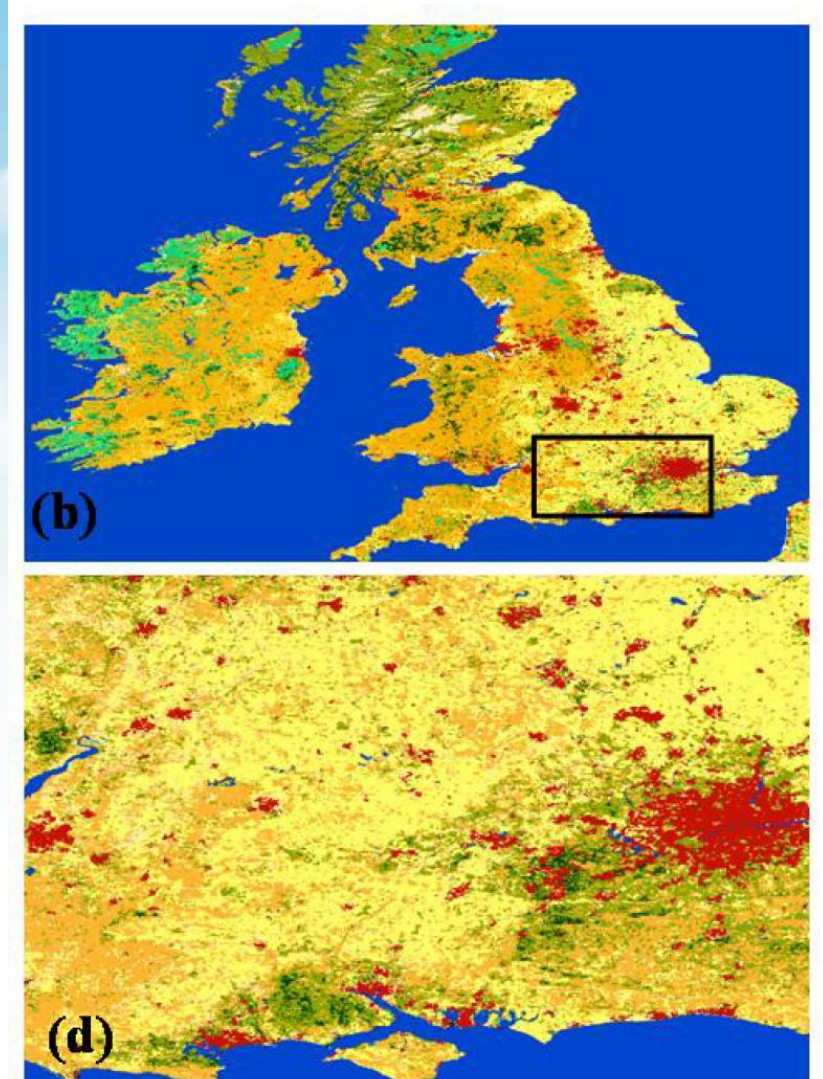
Zhu, X. X., Chen, S., Zhang, F., Shi, Y., and Wang, Y. (2025). GlobalBuildingAtlas: an open global and complete dataset of building polygons, heights and LoD1 3D models. *Earth Syst. Sci. Data* 17, 6647–6668. doi: [10.5194/essd-17-6647-2025](https://doi.org/10.5194/essd-17-6647-2025)
<https://tubvsig-so2sat-vm1.srv.mwn.de/>
<https://huggingface.co/datasets/zhu-xlab/GBA.LoD1>



Данные о типах земельного покрова

Пример: Globcover 2009

Value	Global Globcover legend (level 1)	
11	Post-flooding or irrigated croplands	
14	Rainfed croplands	
20	Mosaic Cropland (50-70%) / Vegetation (grassland, shrubland, forest) (20-50%)	
30	Mosaic Vegetation (grassland, shrubland, forest) (50-70%) / Cropland (20-50%)	
40	Closed to open (>15%) broadleaved evergreen and/or semi-deciduous forest (>5m)	
50	Closed (>40%) broadleaved deciduous forest (>5m)	
60	Open (15-40%) broadleaved deciduous forest (>5m)	
70	Closed (>40%) needleleaved evergreen forest (>5m)	
90	Open (15-40%) needleleaved deciduous or evergreen forest (>5m)	
100	Closed to open (>15%) mixed broadleaved and needleleaved forest (>5m)	
110	Mosaic Forest/Shrubland (50-70%) / Grassland (20-50%)	
120	Mosaic Grassland (50-70%) / Forest/Shrubland (20-50%)	
130	Closed to open (>15%) shrubland (<5m)	
140	Closed to open (>15%) grassland	
150	Sparse (>15%) vegetation (woody vegetation, shrubs, grassland)	
160	Closed (>40%) broadleaved forest regularly flooded - Fresh water	
170	Closed (>40%) broadleaved semi-deciduous and/or evergreen forest regularly flooded - Saline water	
180	Closed to open (>15%) vegetation (grassland, shrubland, woody vegetation) on regularly flooded or waterlogged soil - Fresh, brackish or saline water	
190	Artificial surfaces and associated areas (urban areas >50%)	
200	Bare areas	
210	Water bodies	
220	Permanent snow and ice	



Данные о типах земельного покрова

Data set	Satellite data	Reference & URL	Grid step	Years	Urban class definition	Unit
MODIS Landcover (WRF's default)	MODIS TERRA & AQUA	Sulla-Menashe and Friedl (2018); modis.gsfc.nasa.gov/data/dataproduct/mod12.php	500 m	2001-2020	<i>Urban and built-up lands (at least 30% impervious area)</i>	Binary class
ESA Globcover (COSMO's default)	ENVISAT, MERIS	Bontemps et al. (2011) due.esrin.esa.int/page_globcover.php	300 m	2009	<i>Artificial surfaces and associated areas</i>	Binary class
ESA Climate Change Initiative (ESA CCI)	MERIS, AVHRR, SPOT-VGT, PROBA-V, S3 OLCI	https://www.esa-landcover-cci.org/	300 m	1992-2015	<i>Urban areas</i>	Binary class
Ecoclimap Second Generation	ESA CCI	https://opensource.umr-cnrm.fr/projects/ecoclimap-sg/wiki	300 m	2015(?)	Local Climate Zones (LCZs) according to Stewart & Oke (2012)	LCZ class
Corine LandCover (only EU)	Landsat, Sentinel	https://land.copernicus.eu/en/products/corine-land-cover	100 m	2000-2018	11 urban classes: continuous/discontinuous urban fabric, industry, airports, etc.	Class type
Copernicus Global Land Cover (CGLC)	PROBA-V	Buchhorn et al. (2020); lcviewer.vito.be/ https://zenodo.org/records/3939050	100 m	2015-2019	<i>Land covered by buildings and other man-made structures</i>	Area fraction
GlobeLand30 (GL30)	TM5 ETM+, Landsat, HJ-1, GF-1	Jun et al., (2014); www.globallandcover.com/home_en.html	30 m	2000, 2010, 2020	<i>Artificial Surfaces (All kinds of habitation in urban and rural areas, industrial and mining area)</i>	Binary class
ESA Worldcover	Sentinel 1, 2	Zaranga et al, 2021; viewer.esa-worldcover.org/worldcover/ https://zenodo.org/records/5571936	10 m	2020	<i>Land covered by buildings, roads, etc.</i> Urban green is not included.	Binary class

Больше данных в Google и Zenodo

Специализированные базы данных

❑ Базы данных об искусственных/урбанизированных поверхностях и их типах

Data set	Satellite data	Reference & URL	Grid step	Years	Urban class definition	Unit
Global Man-made Impervious Surface (GMIS)	Landsat	de Colstoun et al., (2017); sedac.ciesin.columbia.edu/data/set/ulandsat-gmis-v1	30 m	2010	<i>man-made imperviousness</i>	Fraction
Global Human Built-up And Settlement Extent (HBASE)		Wang et al., (2017); sedac.ciesin.columbia.edu/data/set/ulandsat-hbase-v1	30 m	2010	<i>urban extent</i>	Fraction
Global Artificial Impervious Areas (GAIA)		Gong et al., (2020); http://data.ess.tsinghua.edu.cn/	30 m	1985-2018	<i>impervious areas</i>	Binary class
Global/continental LCZ maps		Demuzere et al., (2019; 2019; 2022) https://www.wudapt.org/lcz-maps/	100 m		17 Local climate zones (LCZs)	LCZ class

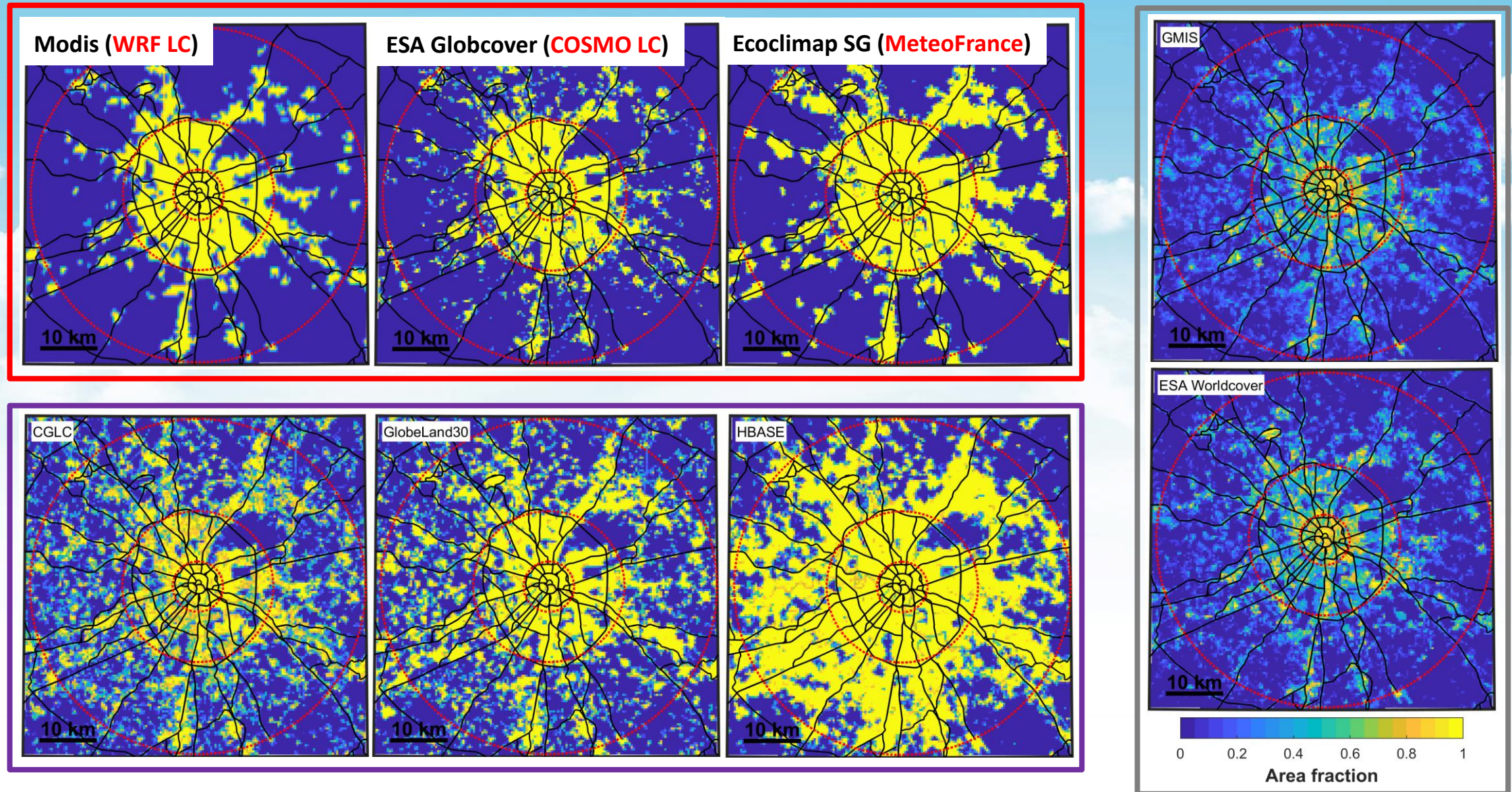
❑ Базы данных об антропогенном потоке тепла

- [Flanner et al., 2009](#) → <https://www.cgd.ucar.edu/projects/ahf>
- [Dong et al, 2017](#) → <https://urbanclimate.tse.ens.titech.ac.jp/2017/02/01/ahe/>
- [Jin et al., 2019](#) → <https://doi.org/10.6084/m9.figshare.c.4182824>
- [Varques et al., 2021](#) → https://figshare.com/articles/dataset/Global_1-km_present_and_future_hourly_anthropogenic_heat_flux/12612458/6

❑ Базы данных о геометрических параметрах застройки

- EU Urban Atlas Building Height, <https://land.copernicus.eu/en/products/urban-atlas/building-height-2012>
- Mapuce (France), <http://mapuce.orbisgis.org/>
- GHSL - Global Human Settlement Layer, <https://ghsl.jrc.ec.europa.eu/index.php>

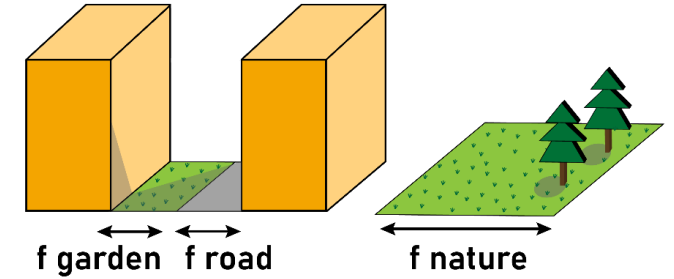
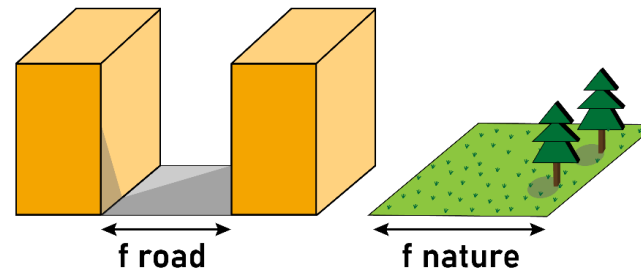
Неопределенность глобальных БД



Что считать городским типом земельного покрова?

Возможные определения:

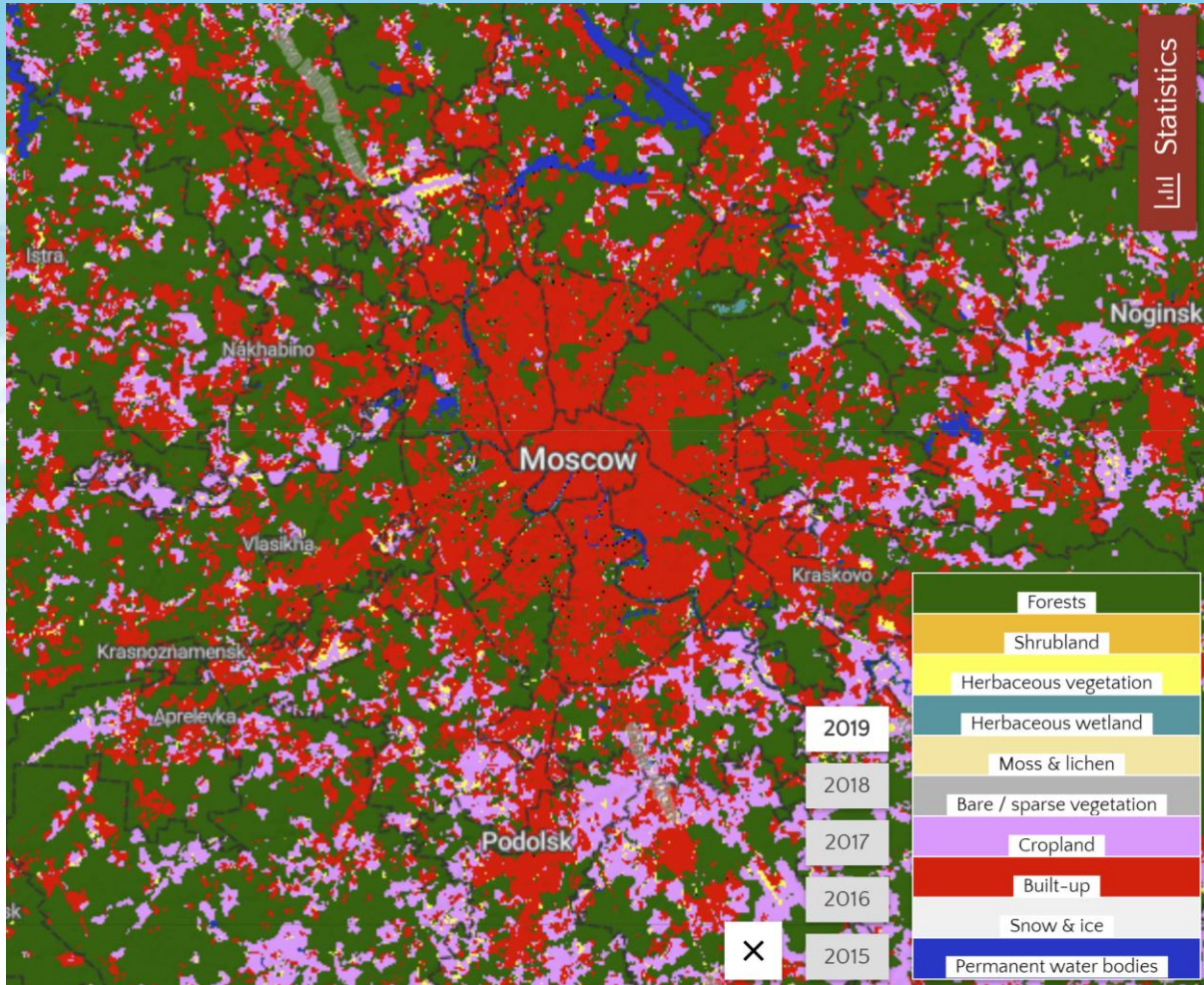
- Запечатанная поверхность (impervious/sealed/paved)
- Застроенная поверхность (built-up)
- Городская поверхность (urban)



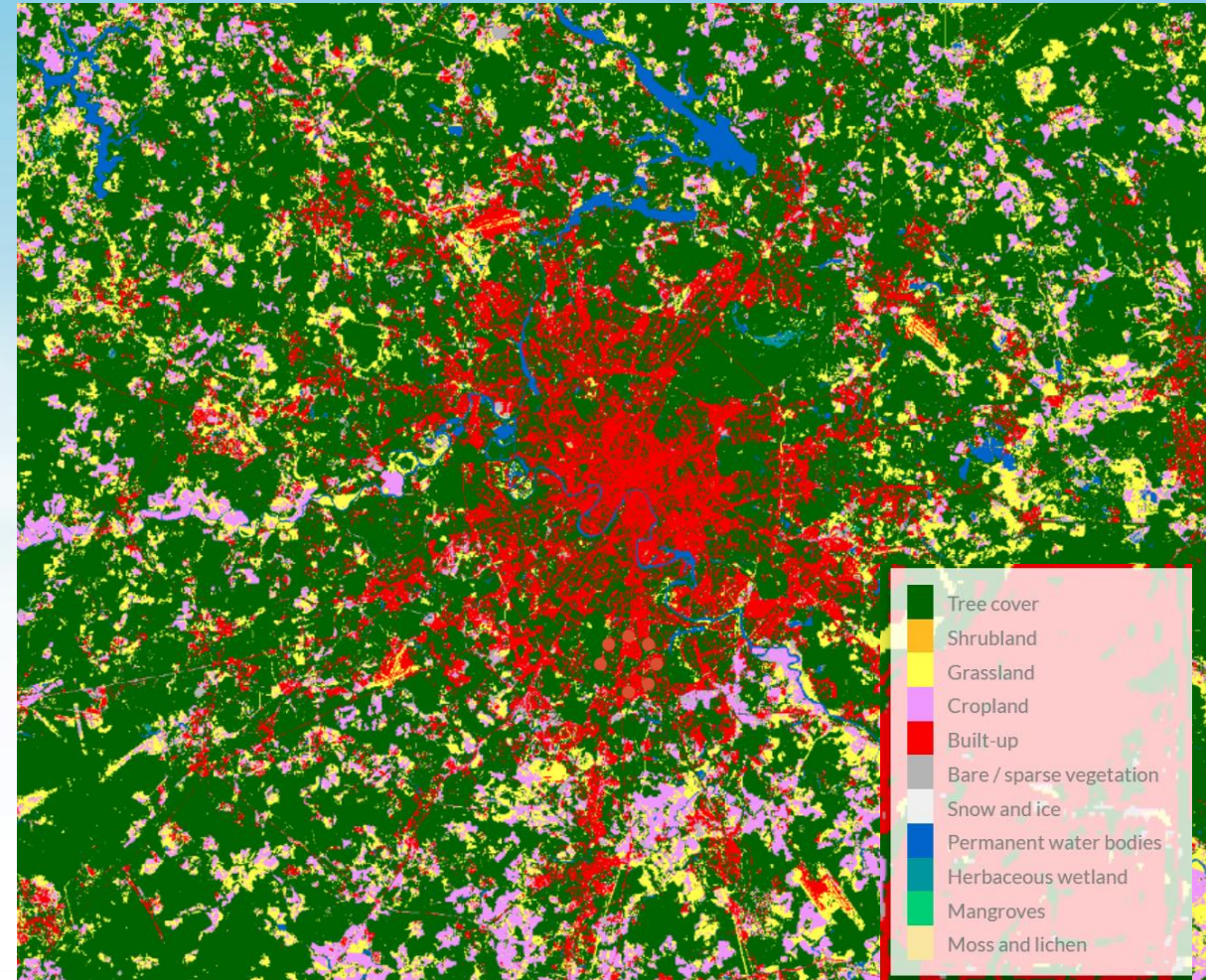
Важно учитывать определение городской поверхности в модели и в источнике входных данных

Сравнение CGLC & Worldcover

CGLC (100 m grid)



Worldcover (10 m grid)



Сравнение CGLC & Worldcover

CGLC (100 m grid)



Worldcover (10 m grid)



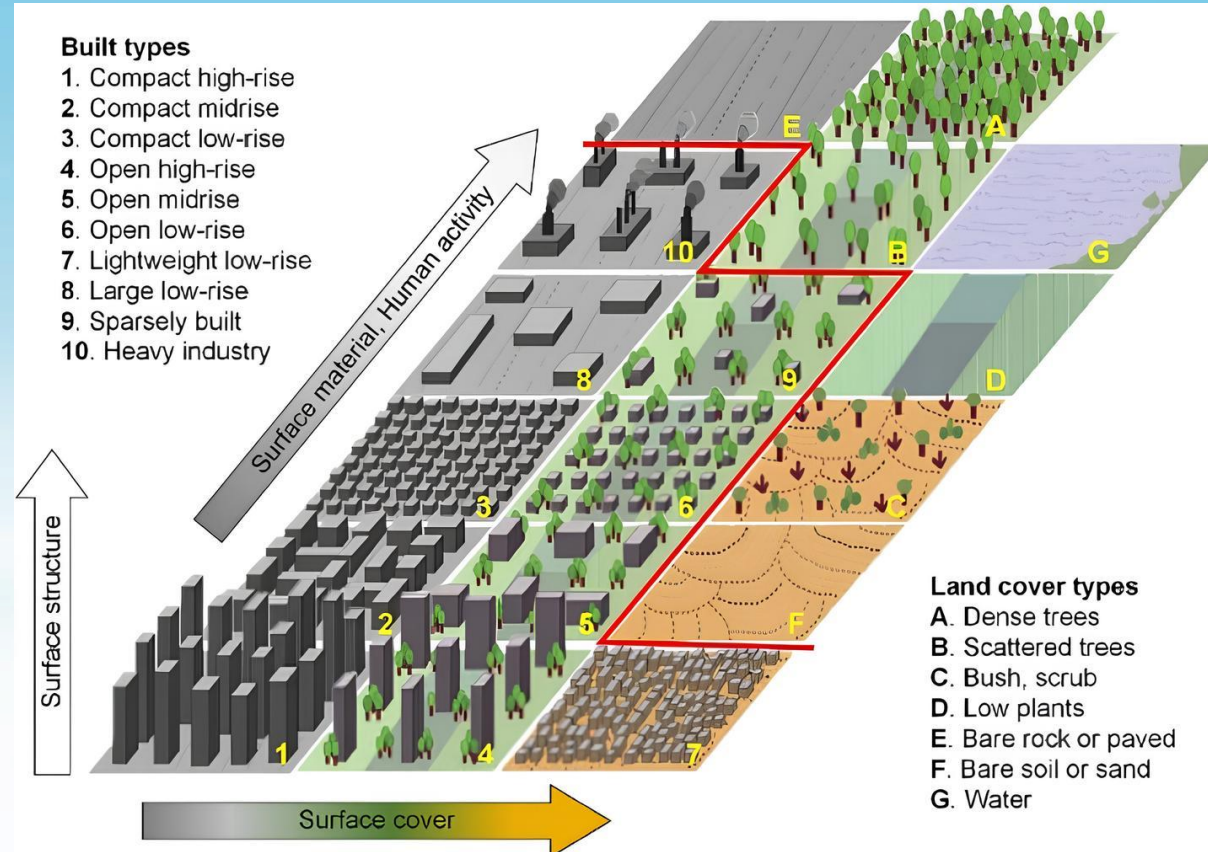
Локальные климатические зоны

- ❑ **Локальные климатические зон (ЛКЗ)** – области с однородным покрытием поверхности, структурой, материалами и особым характером человеческой деятельности, характерным типом взаимодействия с атмосферой (Stewart and Oke, 2012)
- ❑ Краудсорсинговый проект WUDAPT направлен на сбор данных об ЛКЗ (Ching et al., 2018)
- ❑ Локальные карты ЛКЗ для отдельных городов могут быть легко созданы с помощью веб-инструмента LCZ Generator (Demuzere et al., 2021)
- ❑ Доступна глобальная карта ЛКЗ для всей Земли (Demuzere et al., 2022)

World Urban Database

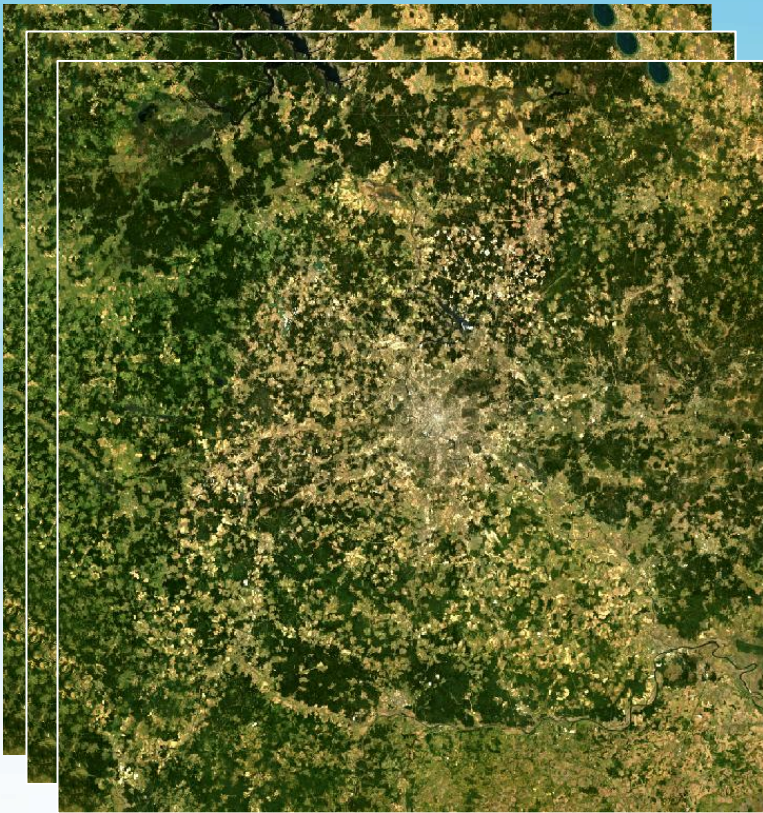
World Urban Database and Access Portal Tools

Ching, J., Mills, G., Bechtel, B., See, L., Feddema, J., Wang, X., ... Theeuwes, N. (2018). WUDAPT: An Urban Weather, Climate, and Environmental Modeling Infrastructure for the Anthropocene. *Bulletin of the American Meteorological Society*, 99(9), 1907–1924. <https://doi.org/10.1175/BAMS-D-16-0236.1>



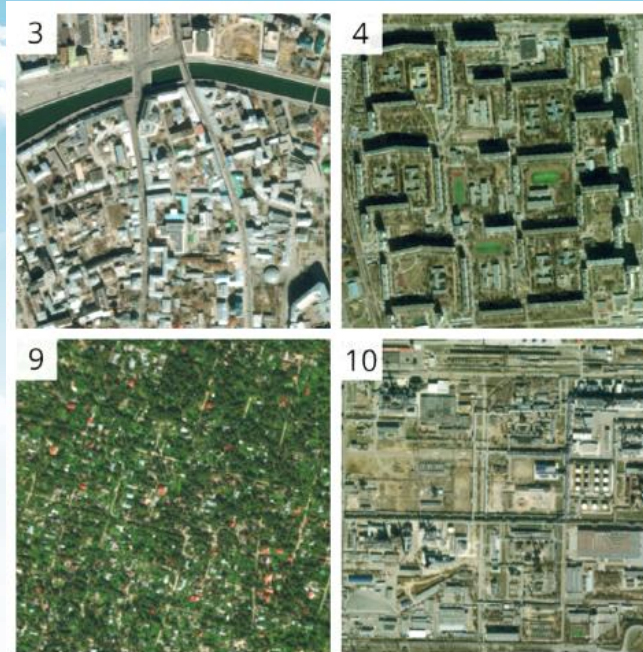
Local climate zone (LCZ)	Sky view factor ^a	Aspect ratio ^b	Building surface fraction ^c	Impervious surface fraction ^d	Pervious surface fraction ^e	Height of roughness elements ^f	Terrain roughness class ^g
LCZ 1	0.2–0.4	> 2	40–60	40–60	< 10	> 25	8
Compact high-rise							
LCZ 2	0.3–0.6	0.75–2	40–70	30–50	< 20	10–25	6–7
Compact midrise							
LCZ 3	0.2–0.6	0.75–1.5	40–70	20–50	< 30	3–10	6
Compact low-rise							

Локальные климатические зоны

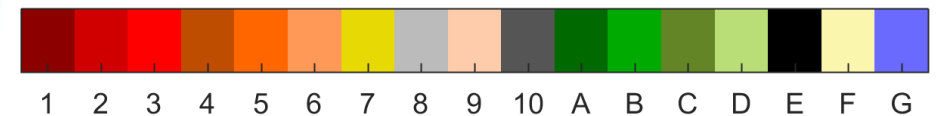
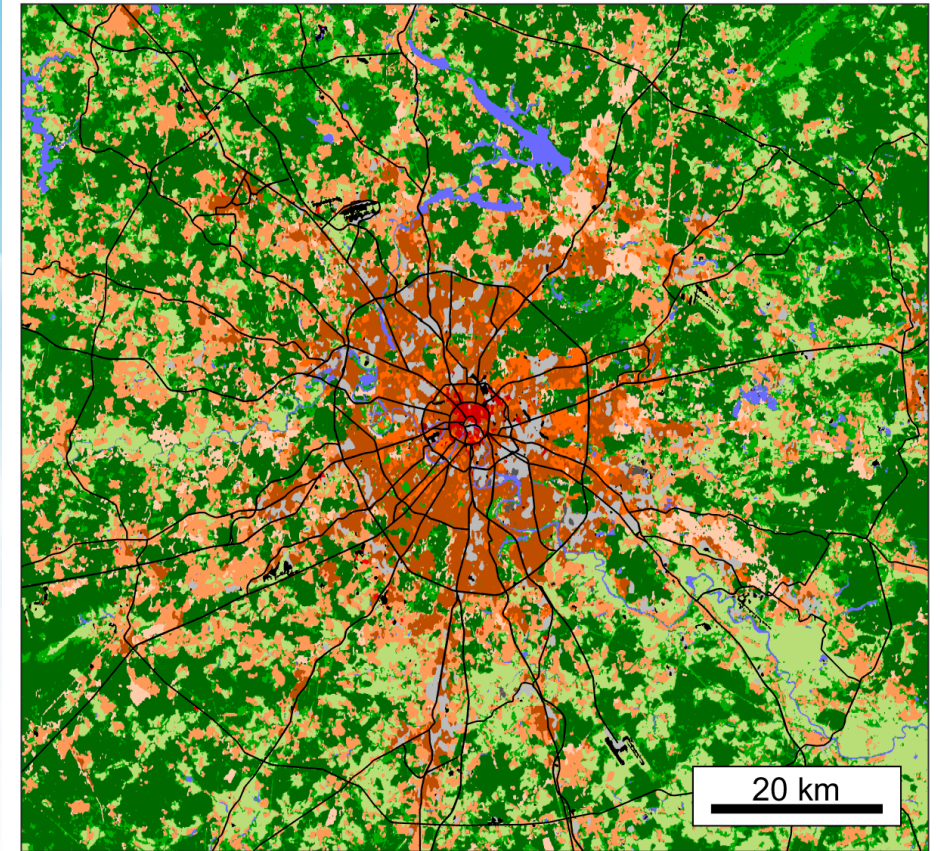


Input features: Landsat 8, Sentinel 1 & 2, Other ...

Training Polygons



Random forest classifier



LCZ Map

Выделение эталонов ЛКЗ

Property	Examples		Property	Examples	
Size Think at the local scale. Individual buildings do not constitute an LCZ. Look for large homogenous areas that are at a minimum 200 m wide at the narrowest point. Use Google Earth's measurement tool to ensure the area is large enough.			Persistence Seasonality Avoid construction sites, as they are likely to change LCZ type during a short period of time.		
Shape Avoid complex shapes, as this can lead to mixed spectral information. Simple block shapes however will maximize the homogeneity of the spectral information and the number of available satellite pixels available within the shape.			The surface characteristics of some LCZ types may depend on the seasonality. Agricultural areas (LCZ D, crops) are a typical example, where land cover may flip from bare soil to cultivated land throughout the year. Use the Google Earth's time slider to explore seasonality.		
Homogeneity If you digitize a training area, the surface characteristics should be similar. In case of doubt, better to digitize different training areas that are homogeneous than one area that is too heterogeneous.			Keep in mind tidal or seasonal waters. Depending on the time of day or season of the year, shorelines or river beds might be dry or contain water.		
Borders Distance Try to keep a minimum distance to other LCZs when classifying. If polygons from different classes are too close to each other, the classifier will receive mixed spectral signals which will affect the quality of the classification. Similarly, do not digitize your training area too close to other land cover(s). Also, avoid precise digitization along road or river segments, features that are often too narrow.			Spatial Distribution Distribute training areas over the entire region of interest, as the same LCZs might differ in their appearance and spectral properties for different parts of your region of interest.		
					

- Можно выполнять в Google Earth
- Инструкция от WUDAPT: <https://www.wudapt.org/digitize-training-areas/>
- Шаблон файла: https://www.wudapt.org/wp-content/uploads/2020/08/WUDAPT_L0_Training_template.kml

Построение карт ЛКЗ

 LCZ Generator

Home Training area submission Available LCZ maps FAQ

Welcome to the LCZ Generator!

Fast and easy Local Climate Zone mapping

Getting started:

1. Read [Demuzere et al. \(2021\)](#) it serves as the primary user guide
2. Download the [Training Area Template](#) kml file
3. Create your Training Areas following the [guidelines](#)
4. Once finished, use the [submission form](#) to submit your file.
5. Fill out the fields in the submission form; fields with an asterisk (*) are required.
▶ Show detailed information
6. Submit the form. If you see a green box appear on the top of the page after clicking the submit button, your submission was successful and will be processed. If a red box appears, there was a problem with your Training Area file. Check out the [FAQ](#) for more information.
7. You will be notified via email once the processing has finished. Depending on the current load of the system it should take ~20 minutes.
8. After you received the email, your submission is also available in the [submission table](#).

Submit your Training Area

Show generated LCZ maps

Please cite the tool using:

Demuzere, M., Kittner, J., Bechtel, B. (2021). LCZ Generator: a web application to create Local Climate Zone maps. *Frontiers in Environmental Science* 9:637455. <https://doi.org/10.3389/fenvs.2021.637455>

<https://lcz-generator.rub.de/>

Submitted LCZ Maps

LCZ maps generated with the LCZ Generator. Only the best result per author and city is displayed in this table.
The authors are responsible for the quality of the LCZ map. A high OA (Overall Accuracy) does not mean that the map is correct. See the [FAQ page](#) for more details.

Overall Accuracy $\geq$ 0

Show 10 entries

[Download Selected](#) [Show all submissions](#)

Search:

☐	City	Country	Continent	Date Submitted	Author	Accuracy	
☐	Nizhny Novgorod	Russian Federation	Europe	2021-04-14 15:01:04		0.74	Show Factsheet
☐	Nobosibirsk	Russian Federation	Europe	2022-05-12 04:57:56		0.67	Show Factsheet
☐	Novosibirsk	Russian Federation	Asia	2021-12-07 02:28:18	Aleksander Gochakov	0.71	Show Factsheet
☐	Novosibirsk	Russian Federation	Europe	2022-05-11 10:40:09		0.69	Show Factsheet
☐	Saint Petersburg	Russian Federation	Asia	2021-04-14 13:23:34	Teresa Mansheim	0.73	Show Factsheet
☐	Yakutsk	Russian Federation	Europe	2021-04-14 15:01:28		0.86	Show Factsheet

Глобальная карта ЛКЗ

Earth Syst. Sci. Data, 14, 3835–3873, 2022

<https://doi.org/10.5194/essd-14-3835-2022>

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29 Aug 2022

A global map of local climate zones to support earth system modelling and urban-scale environmental science

Matthias Demuzere¹, Jonas Kittner¹, Alberto Martilli², Gerald Mills³, Christian Moede¹, Iain D. Stewart⁴, Jasper van Vliet⁵, and Benjamin Bechtel¹

¹Urban Climatology Group, Department of Geography, Ruhr-University Bochum, Bochum, Germany

²Environmental Department, CIEMAT, Madrid, Spain

³School of Geography, University College Dublin, Dublin, Ireland

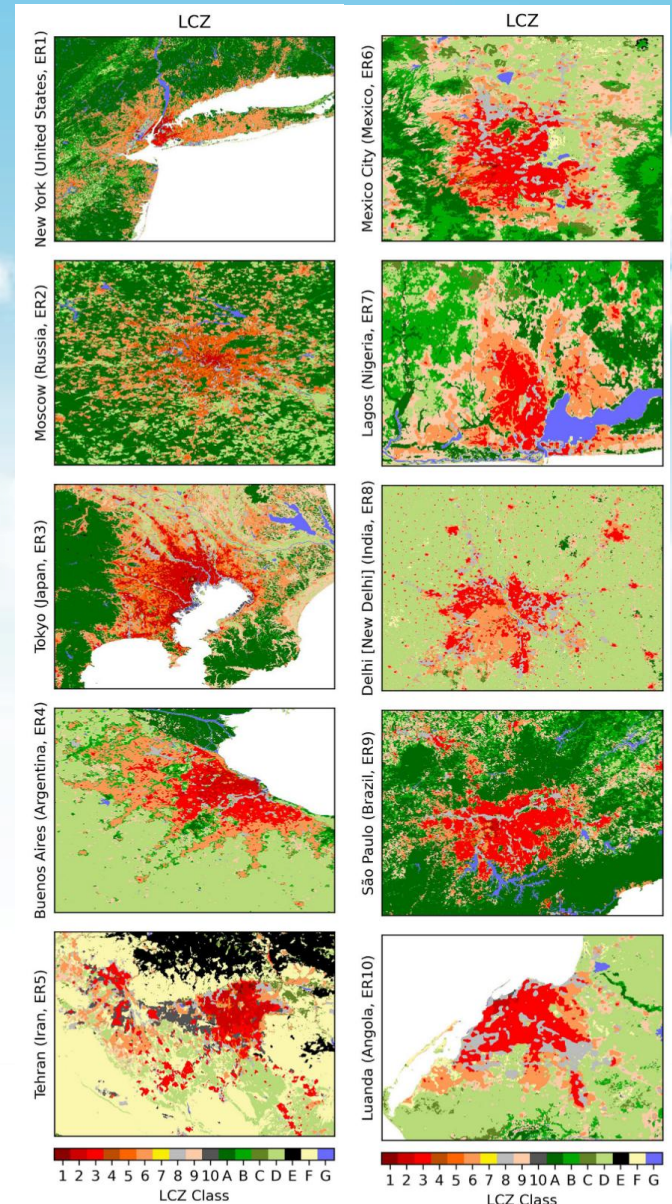
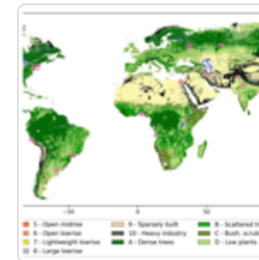
⁴Global Cities Institute, University of Toronto, Toronto, Ontario, Canada

⁵Institute for Environmental Studies, Vrije Universiteit Amsterdam, De Boelelaan 1085, 1081, HV, Amsterdam, the Netherlands

Correspondence: Matthias Demuzere (matthias.demuzere@rub.de)

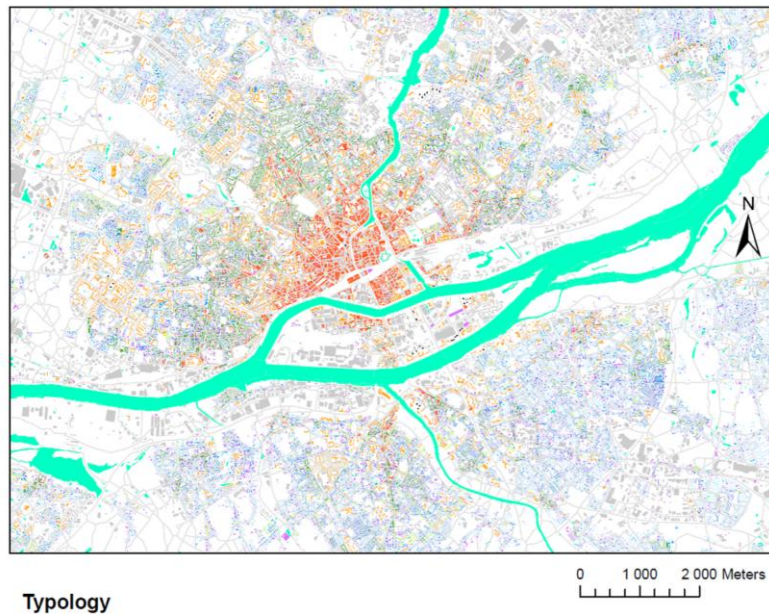
<https://zenodo.org/records/8419340>

<https://lcz-generator.rub.de/global-lcz-map>



Специализированные базы данных

MAPUCE (Франция)



NUDAPT (США)

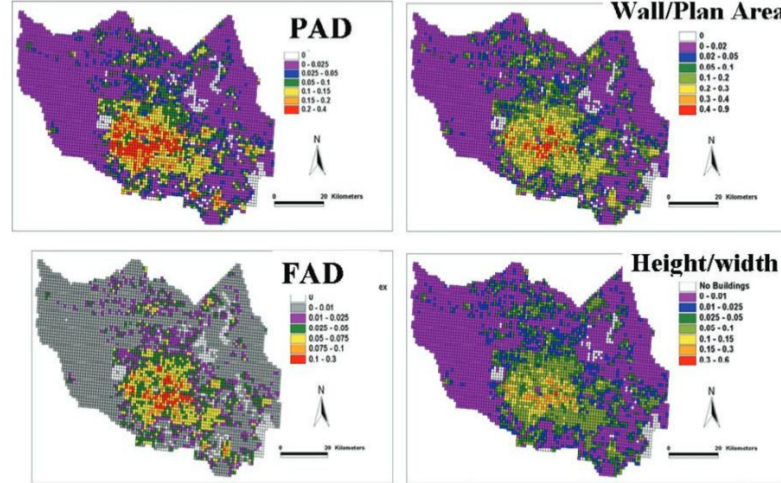
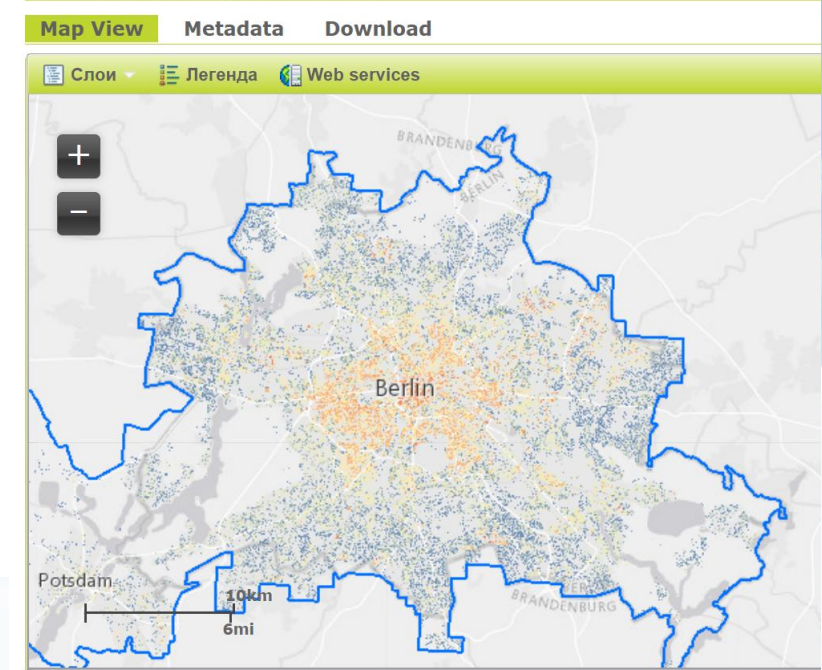


FIG. 2. Selected UCPs derived for 1-km² cells for Harris County, TX, as used in the urbanized MM5 system. PAD is plan area density, and FAD is frontal area density of the buildings in each cell. Note that each cell has a unique combination of UCPs.

EU Urban Atlas

Building Height 2012



Специализированные базы данных

GHSL - Global Human Settlement Layer

Open and free data and tools for assessing the human presence on the planet



GHS-BUILT-S

GHS-BUILT-S New release

The spatial raster dataset depicts the distribution of built-up surfaces, expressed as number of square metres. The data report about the total built-up surface and the built-up surface allocated to dominant non-residential (NRES) uses.



GHS-BUILT-H

GHS-BUILT-H New release

The spatial raster dataset depicts the spatial distribution of the building heights as extracted from the filtering of a composite of global digital elevation models (DEM) and the filtering of satellite imagery using linear regression techniques.

<https://ghsl.jrc.ec.europa.eu/index.php>

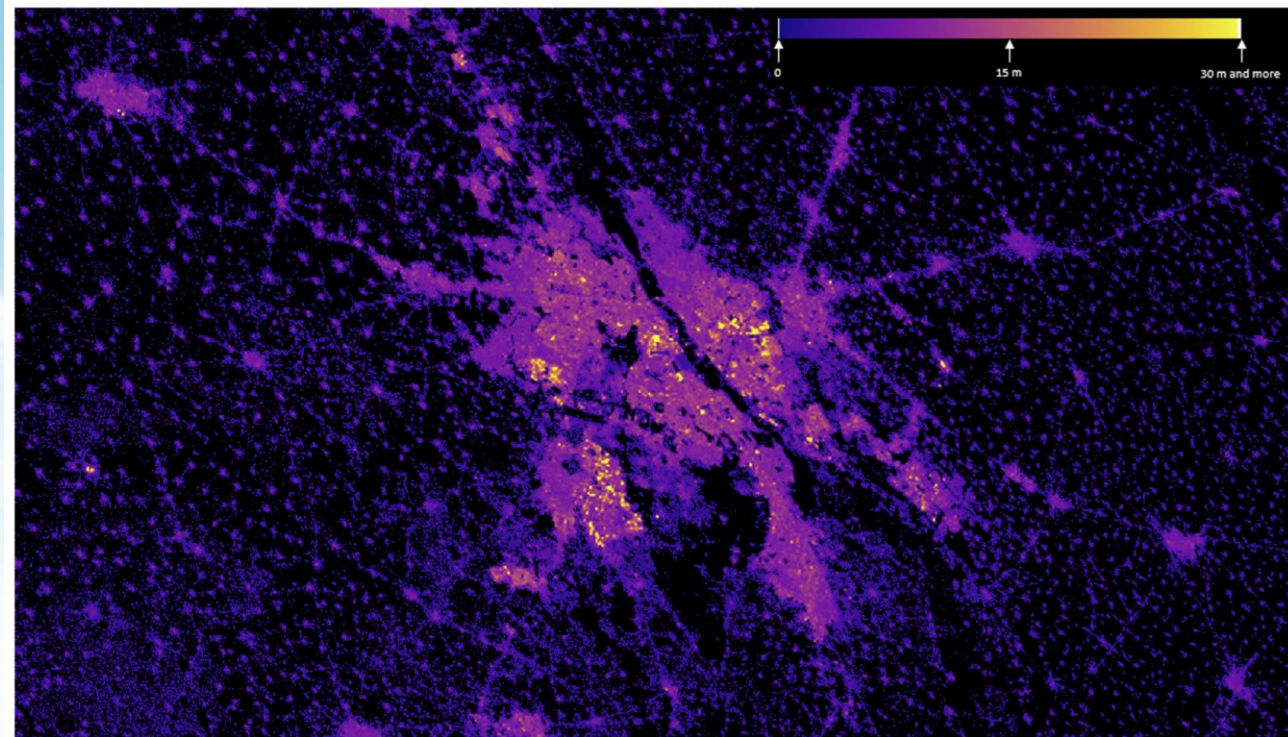


Figure 12 - Average building height (ANBH 100m) estimates in Delhi (India).

Специализированные базы данных

World Urban Morphological Parameters Open Dataset (WUMPOD)

Patel, Pratiman¹ ; Roth, Matthias¹ 

Show affiliations

WUMPOD is available with a spatial resolution of 500 m in Cloud Optimized GeoTIFF (COG) format and WRF-WPS format for easy access and utilization within the WRF modelling system. The COG format uses "Lempel–Ziv–Welch (LZW)" compression, overviews are embedded, and contains separate files of the following urban morphological parameters; MH (mean building height), STDH (standard deviation of building height), HGT (area weighted mean building height), LP (building plan area fraction), LB (building surface to plan area fraction), LF (frontal aspect ratio for cardinal directions of 0, 45, 90, and 135 degrees), LC (complete aspect ratio), H2W (height to width ratio), HI [0-75 m] (percentage distribution of building heights at every 5 m up to 75 m), ZOM, ZDM (Roughness length and displacement height according to Macdonald et al. (1998) for 0, 45, 90, and 135 directions), urban fraction (from WorldCover 2021) and missing building fraction. The WRF-WPS binary dataset contains additional variables, i.e., ZOS, ZDS (Roughness length and displacement height according to Grimmond and Oke (1999)), ZOR, and ZDR (Roughness length and displacement height according to Raupach (1994) for 0, 45, 90, and 135 directions), but it does not contain missing and urban fraction variables. Please read WRF-WPS readme file to use it in the WRF simulations. It is important to note that WRF-WPS binary, when uncompressed is approximately 630 GB. It is advised that user check disk space before uncompressing the file.

The urban fraction in WRF-WPS format is available at <https://zenodo.org/records/7298393>

Abstract (English)

Urbanization is associated with the physical growth of cities, which disrupts the local ecosystem through changes to the energy balance and outputs of waste heat, greenhouse gases or air pollutants. Over 50% of the global population resides in urban areas characterized by particular urban climates, making them increasingly vulnerable to climate change-related risks. To project future conditions or address climate adaptation and mitigation challenges, global climate models are starting to incorporate urban physics while regional and high-resolution climate models have made progress in utilizing urban canopy models representing the urban surface. However, the lack of comprehensive global urban morphology data limits these efforts and accuracy of predictions. Obtaining global data, such as building height and footprint, is a complex and computationally costly process, and existing community initiatives such as NUDAPT, MapUCE, and deep learning approaches are limited to regional coverage. The present dataset aims to create a high-resolution (500 m) global database using only open data sources to support global climate modelling and improve urban-scale climate projections.

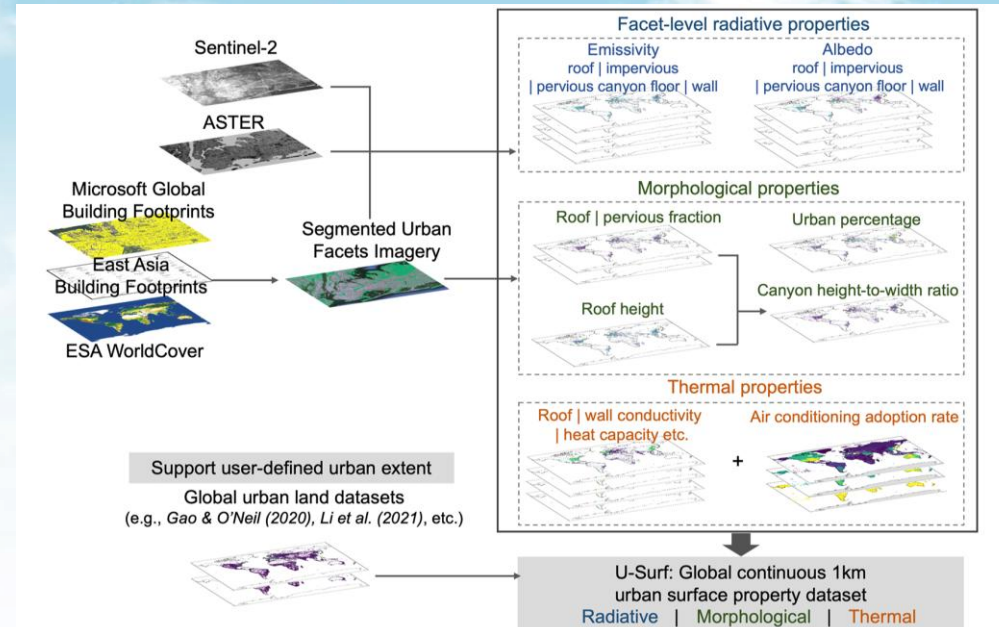
<https://zenodo.org/records/10039127>

Специализированные базы данных

Category	Urban Parameters	U-Surf	CLMU
Radiative	Roof Emissivity	- Source: 100m ASTER v3 emissivity product (Hulley et al., 2015) - Time span: static, representing 2000-2008 - Spatial resolution: 1km	- Source: local building codes, municipal documentation, literature, satellite imageries (Jackson et al., 2010; Oleson and Feddema, 2020) - Time span: 1966-2007 - Spatial resolution: Regional-level, density-class-specific
	Impervious Canyon Floor Emissivity		
	Pervious Canyon Floor Emissivity		
	Wall Emissivity*		
	Roof Albedo	- Source: 10m Sentinel2 albedo product (Lin et al., 2022) and narrow-to-broadband algorithm (Bonafoni & Sekertekin, 2020) - Time span: 2019-2022 - Spatial resolution: 1km	
	Impervious Canyon Floor Albedo		
	Pervious Canyon Floor Albedo		
	Wall Albedo*		
Morphological	Building Height	- Source: 3D-GloBFP (Che et al., 2024), 3D building structure (Li et al., 2022) - Time span: 2014-2021 - Spatial resolution: 1km	- Source: local building codes, municipal documentation, literature, satellite imageries (Jackson et al., 2010; Oleson and Feddema, 2020) - Time span: 2000-2007 - Spatial resolution: Regional-level, density-class-specific
	Canyon Height-to-width Ratio	- Source: infinite canyon street model (Masson et al., 2020) - Time span: 2014-2021 - Spatial resolution: 1km	
	Roof Fraction	- Source: Microsoft global building footprints (Microsoft, 2024), East Asia building footprints (Shi et al., 2024) - Time span: 2014-2021 - Spatial resolution: 1km	
	Pervious Canyon Floor Fraction	- Source: 10m ESA Worldcover v200 (Zanaga et al., 2022) - Time span: 2021-2022 - Spatial resolution: 1km	
	Urban Percentage	- Source: building footprints (Microsoft, 2024, Shi et al., 2023) and ESA (Zanaga et al., 2022) - Time span: 2014-2021 - Spatial resolution: 1km	- Source: LandScan global population database (Bright et al., 2005) - Time span: 2004 - Spatial resolution: 1km
Thermal	Air Conditioning Penetration Rate	- Source: global AC penetration rate (Li et al., 2024) - Time span: present-day, loosely defined as 2010-2020 - Spatial resolution: national and sub-national level	AC penetration rate is not explicitly modeled in CLMU as of CLM5 (Oleson and Feddema, 2020)
	Number of Impervious Canyon Layers	- Source: local building codes, municipal documentation, literature, satellite imageries (Jackson et al., 2010; Oleson and Feddema, 2020) - Time span: 1966-2007 - Resolution: Regional-level, density-class-specific**	
	Roof Thickness		
	Wall Thickness		
	Minimum Interior Building Temperature		
	Maximum Interior Building Temperature		
	Roof Thermal Conductivity		
	Impervious Canyon Thermal Conductivity		
	Wall Thermal Conductivity		
	Roof Volumetric Heat Capacity		
	Impervious Canyon Volumetric Heat Capacity		
	Wall Volumetric Heat Capacity		

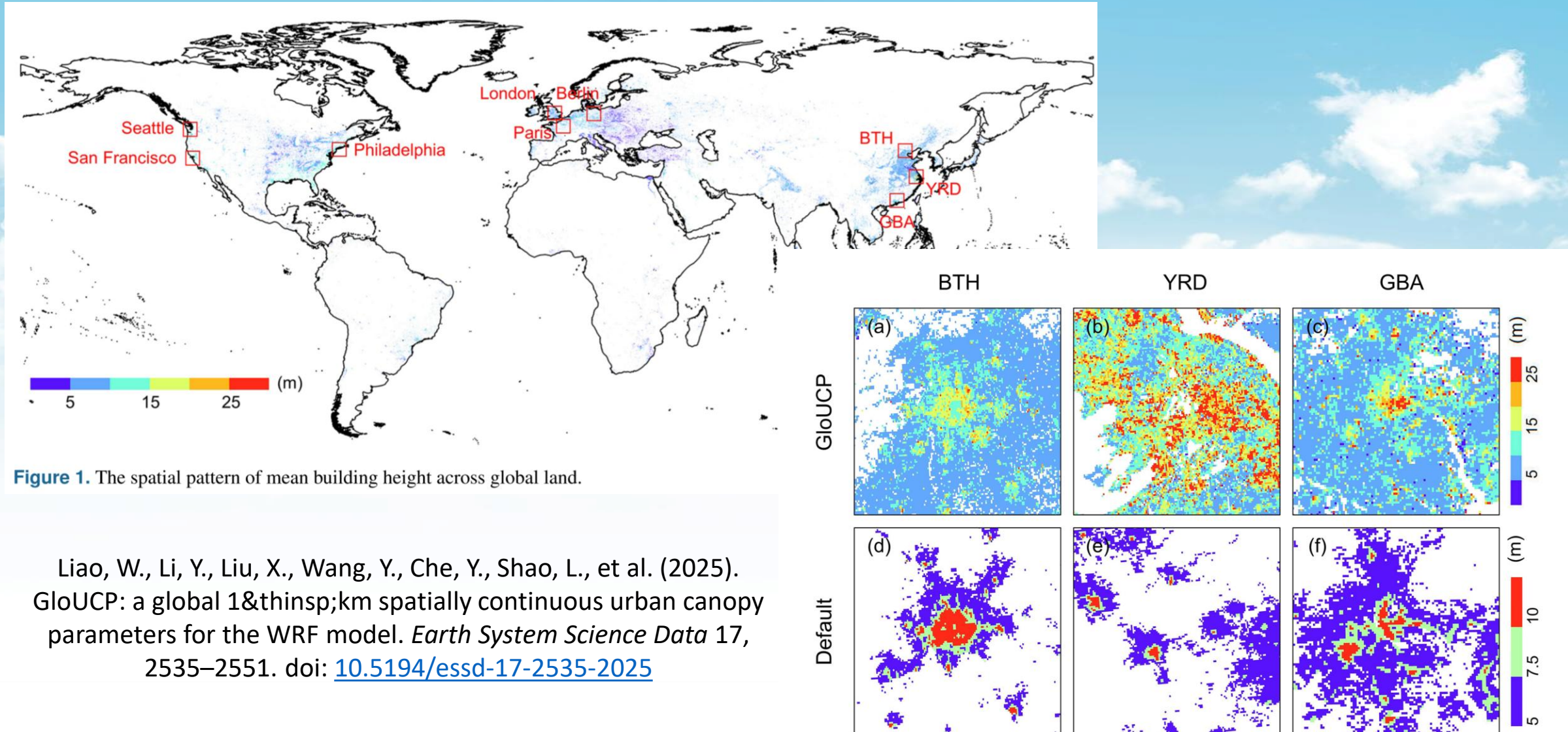
U-Surf: A global 1km spatially continuous urban surface property dataset for kilometer-scale urban-resolving Earth system modeling

Yifan Cheng¹, Lei Zhao^{1,2,3}, Tirthankar “TC” Chakraborty⁴, Keith Oleson⁵, Matthias Demuzere⁶, Xiaoping Liu⁷, Yangzi Che⁷, Weilin Liao⁷, Yuyu Zhou⁸, Xinchang “Cathy” Li¹



<https://doi.org/10.5281/zenodo.11247599>

Специализированные базы данных



Домашнее задание №4

Пространственные данные для моделирования городского климата

Часть 1. Локальные климатические зоны (ЛКЗ)

- Выделите эталоны ЛКЗ для выбранного города по [рекомендациям WUDAPT](#). Используйте при этом собственные экспертные знания, веб-карты, панорамы улиц и пр.
- Постройте региональную карту ЛКЗ используя LCZ Generator (<https://lcz-generator.rub.de/>)
- Рассчитайте доли площади, занимаемые различными типами ЛКЗ в пределах области исследования, сравните их с глобальной картой ЛКЗ ([Demuzere et al., 2022](#); <https://zenodo.org/records/8419340>)

Часть 2. Векторные данные о застройке

- Скачайте данные о зданиях OpenStreetMap или Overture Maps для одного из районов выбранного города. Эта территория должна включать несколько различных ЛКЗ.
- Постройте карту этажности зданий.
- Рассчитайте долю зданий с известной высотой и этажностью.
- *Сравните данные Overture Maps и OpenStreetMap по полноте данных о застройке.

Часть 3. Глобальные базы данных

- Скачайте для выбранного города данные как минимум двух различных глобальных баз данных о типах земельного покрова и/или специализированных баз данных.
- Сравните пространственное распределение урбанизированных территорий по этим базам данных друг с другом и с картой ЛКЗ (постройте серию единообразных карт). Сделайте выводы о репрезентативности выбранных баз данных.
- *Скачайте данные как минимум одной глобальной БД по высоте/этажности зданий (U-Surf, GHSL, WUMPOD). Сравните с картой ЛКЗ и, на примере отдельного района, с векторными данными (OSM/Overture)

Технические вопросы

Работа с растровыми данными

Библиотека rasterio

File connection

```
src = rasterio.open('data/srtm.tif')
src

<open DatasetReader name='data/srtm.tif' mode='r'>
```

dict Metadata

```
src.meta

{'driver': 'GTiff',
 'dtype': 'uint16',
 'nodata': 65535.0,
 'width': 465,
 'height': 457,
 'count': 1,
 'crs': CRS.from_epsg(4326),
 'transform': Affine(0.0008333333332777796, 0.0, -113.23958321278403,
 0.0, -0.0008333333332777843, 37.512916763165805)}
```

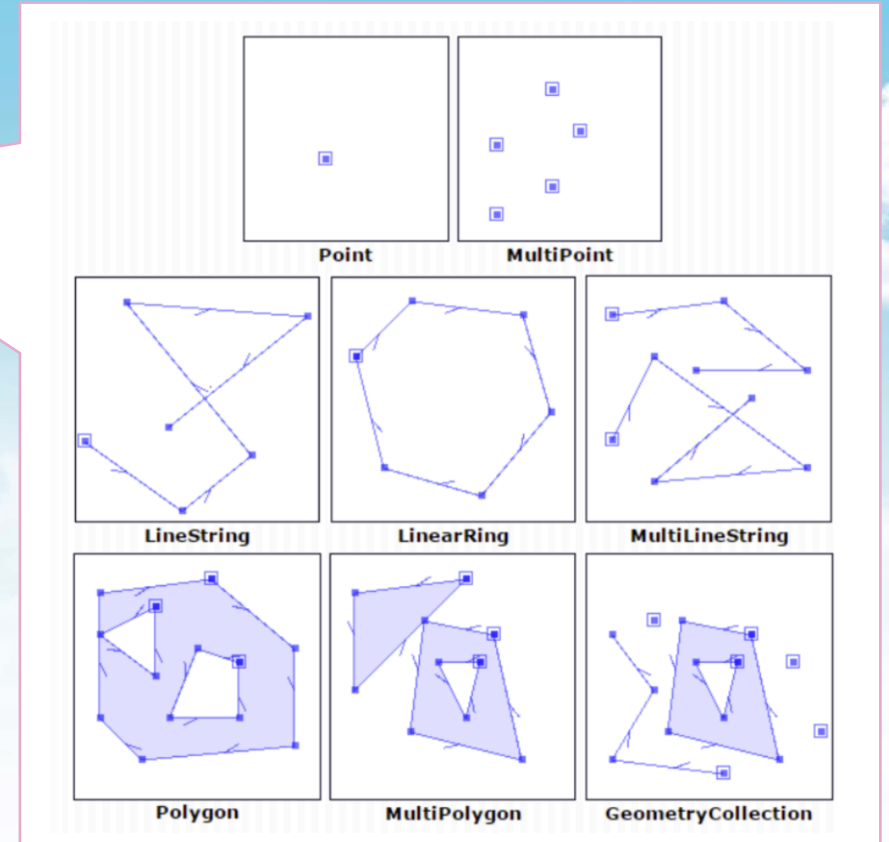
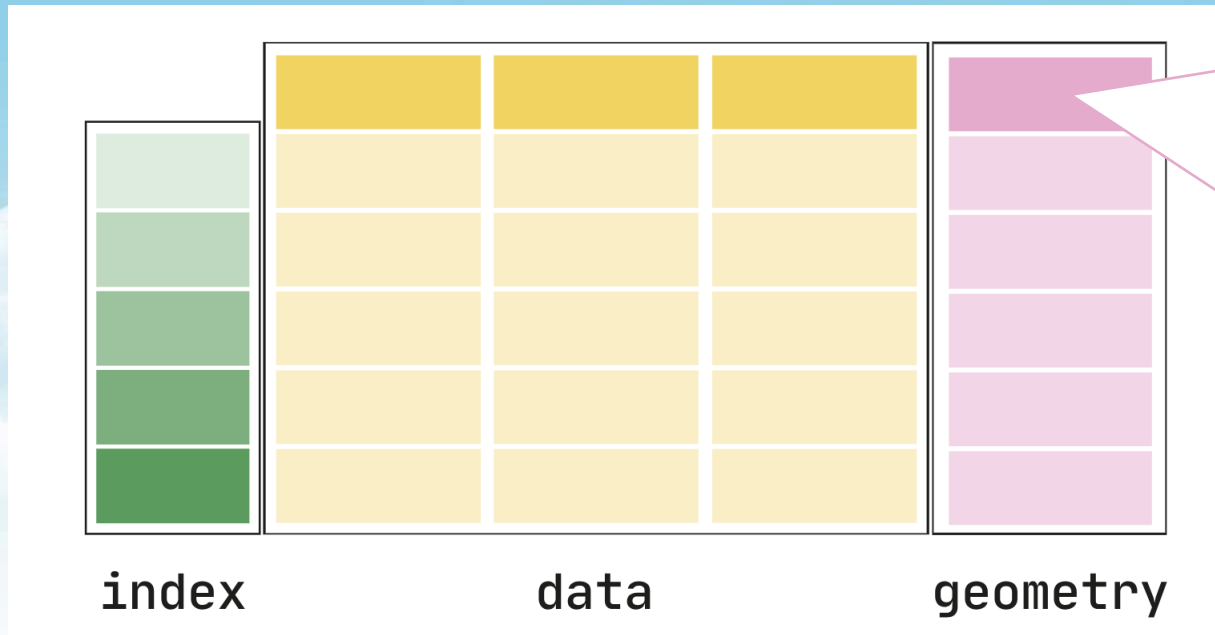
ndarray Values

```
src.read(1)

array([[1728, 1718, 1715, ..., 2654, 2674, 2685],
       [1737, 1727, 1717, ..., 2649, 2677, 2693],
       [1739, 1734, 1727, ..., 2644, 2672, 2695],
       ...,
       [1326, 1328, 1329, ..., 1777, 1778, 1775],
       [1320, 1323, 1326, ..., 1771, 1770, 1772],
       [1319, 1319, 1322, ..., 1768, 1770, 1772]], dtype=uint16)
```

<https://rasterio.readthedocs.io/en/stable/#>

Работа с векторными данными (OSM)



 Shapely +  pandas =  GeoPandas

<https://geopandas.org/en/stable/>